

Faculty of Science & Technology
Programme Structure & Syllabus
For
Third Year
Bachelor of Technology
Computer Engineering
Programme Code: BTECCE
Pattern 2023
< With Effect from Academic Year 2025-26 >

	Authority	Date
Proposed by	Board of Studies in Computer Engineering	
Approved by	Academic Council, Vishwakarma University, Pune	

Issued by

Chairman – BoS

Dean of Faculty

Director, IQAC

Title: Programme Structure and Syllabus

Form No: IQAC-101

Vision of the University	
Emerge as a Premier University Recognized Internationally for Excellence in Education, Research and Innovation	
Mission of the University	
VU-M1	To impart contemporary transformative education through research and innovation
VU-M2	To develop competent leaders-professionals for life and livelihood
VU-M3	To co-create human and socio-economic capital par excellence
VU-M4	To inculcate life skills and holistic culture appreciating morals and ethics
Values of the University	
Excellence	Transparency
Innovation	Sustainability
Diversity	Responsibility
Adaptability	Compassion

Vision of the Department of Computer Engineering	
To create an intellectual and academically rich environment for careers in Computing Education and Research to fulfill global needs.	
Mission of the Department of Computer Engineering	
M1	To impart modern computing education through novel methods and research.
M2	To prepare the students for life-long learning in pursuit of excellence.
M3	To create technically competent human in socio-economic domain.
M4	To promote aspiring ethically conscious engineers demonstrating sustainable entrepreneurship and professional maturity in social context.
Values of the Department of Computer Engineering	
Pursuing Excellence	Ethics & Social Responsibility
Nurturing Talent	Collaborative Approach
Career Architect	Sustainable Growth Transparency & Accountability
Innovative Outlook	Freedom of Expression

Mapping of Mission Statement of Department to University Mission Statement

Mission Statement	VU-M1	VU-M2	VU-M3	VU-M4
M1	3	3	2	2
M2	1	2	2	3
M3	2	3	2	2
M4	1	1	2	3

Affinity Level: 1- Slight, 2- Moderate, 3-Substantial

Programme Educational Objectives (PEOs)

PEO No.	Statement
PEO1	To prepare the students to evolve into a professional and committed technology workforce, by providing them with a global educational platform with innovative practices.
PEO2	To provide a sufficient mathematical and computing theory knowledge base, leading to practical and long lasting computer engineering solutions.
PEO3	To ingrain problem solving skills and sound engineering principles in students with the help of modern tools and techniques.
PEO4	To impart in the students professional and ethical practices based on standard guidelines to acquire the right attitude and aptitude for the benefit of society.
PEO5	To guide ambitious students for higher education and lead them to avenues of entrepreneurship in emerging areas of computing.

Mapping of Mission Statement of Department to PEOs

Mission Statement	PEO1	PEO2	PEO3	PEO4	PEO5
M1	3	3	3	1	2
M2	2	2	3	3	2
M3	3	3	2	2	1
M4	2	1	2	3	2

Affinity Level: 1- Slight, 2- Moderate, 3-Substantial

Programme Outcomes (POs)

PO No.	Statement
PO1	Apply the knowledge of engineering and science fundamentals to the solution of complex engineering problems.
PO2	Identify, formulate and analyze complex engineering problems to create solutions using the First principles of engineering sciences and mathematics.
PO3	Design solutions for design system components that meet the specified needs for the benefit of society.
PO4	Apply research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
PO5	Create appropriate techniques and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
PO6	Apply reasoning using contextual knowledge to assess the needs of society and understand The responsibilities relevant to the professional engineering practice.
PO7	Understand the impact of professional engineering solutions as applied to the environment, And demonstrate the knowledge for sustainable development.
PO8	Adopt ethical practices as applied to the professional world to execute responsibilities and adhere to norms of the engineering profession.
PO9	Execute professional functions effectively as an individual, as well as a leader or member in Diverse multidisciplinary teams.
PO10	Communicate effectively with the engineering community and with society in solving complex problems in terms of being able to comprehend and write effective reports, make Effective presentations, as well as execute and receive clear instructions.
PO11	Demonstrate an ability to lead projects and build cost models in an interdisciplinary professional Setting
PO12	Develop learning mechanisms and inculcate the ability to prepare for lifelong learning in the context of technological change.

Mapping of PEOs to POs

PEO Number	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
PEO1	2	3	2	3	1	2	3	1	1	-	-	2
PEO2	3	3	2	3	1	1	2	1	-	1	1	1
PEO3	2	2	2	2	3	1	3	1	-	1	-	2
PEO4	1	1	1	1	1	1	1	3	3	2	2	-
PEO5	2	2	2	2	1	1	1	1	1	-	2	1

Affinity Level: 1- Slight, 2- Moderate, 3-Substantial

Programme Specific Outcomes (PSOs)

PSO No.	Statement
PSO1	Imbibe foundational knowledge of computing theory, algorithms and programming paradigms to create innovative, scientific solutions for complex computing challenges.
PSO2	Prepare for the latest trends in industry by adapting to various computing platforms, architectures and changing software paradigms.
PSO3	Evolve new design ideas for real world problems using appropriate mechanisms, design patterns, modeling methods and modern tools.
PSO4	Inculcate research attitude and aptitude in students using techniques and standards as applied to technology migration and evolution.

Mapping of PEOs to PSOs

PEO Number	PSO1	PSO2	PSO3	PSO4
PEO1	2	3	1	2
PEO2	3	2	1	3
PEO3	3	2	3	2
PEO4	2	2	1	1
PEO5	2	2	1	2

Affinity Level: 1- Slight, 2- Moderate, 3-Substantial

Programme Structure

Faculty	<i>Science and Technology</i>	Pattern	2023
Department	<i>Computer Engineering</i>	Date (w.e.f.)	01/07/2025
Programme	<i>B.Tech. Computer Engineering</i>		

Semester –V Pattern 2023									
Course Type	Course Code	Course Name	Teaching Scheme(Hours/Week)			Credit	Examination Scheme and Marks		
			L	T	P	C	CI E	ES E	Total
PCC	BTECCE23501	Operating System	3	0	0	3	60	40	100
PCC	BTECCE23502	Operating System Lab	0	0	2	1	15	10	25
PCC	BTECCE23503	Computer Network	3	0	0	3	60	40	100
PCC	BTECCE23504	Computer Network Lab	0	0	2	1	15	10	25
PCC	BTECCE23505	Design and Analysis of Algorithms	2	0	0	2	50	0	50
PEC	BTECCE23506A	Compiler Design	3	0	0	3	60	40	100
PEC	BTECCE23506B	Embedded System Design	3	0	0	3	60	40	100
PEC	BTECCE23507A	Compiler Design Lab	0	0	2	1	15	10	25
PEC	BTECCE23507B	Embedded System Design Lab	0	0	2	1	15	10	25
MC	MT23CE03A	Software Engineering	3	0	0	3	60	40	100
OE	OE23CE03	Software Project Management	3	1	0	4	60	40	100
TOTAL			17	1	6	21	395	230	625
			21						
Instructions, if any: 1 Theory/Tutorial Hour = 1 Credit, 2 Practical hours = 1 Credit									

Hons	BTECCE23H508	Network Security	3	0	0	3	60	40	100
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Faculty	<i>Science and Technology</i>	Pattern	2023
Department	<i>Computer Engineering</i>	Date (w.e.f.)	01/07/2025
Programme	<i>B.Tech. Computer Engineering</i>		

Semester –VI Pattern 2023									
Course Type	Course Code	Course Name	Teaching Scheme(Hours/ Week)			Credit	Examination Scheme and Marks		
			L	T	P	C	CIE	ESE	Total
PCC	BTECCE23601	Artificial Intelligence	3	0	0	3	60	40	100
PCC	BTECCE23602	Artificial Intelligence Lab	0	0	2	1	15	10	25
PCC	BTECCE23603	Statistical Data Analysis and Visualization	3	0	0	3	60	40	100
PCC	BTECCE23604	Statistical Data Analysis and Visualization Lab	0	0	2	1	15	10	25
PEC	BTECCE23605A	Machine Learning	3	0	0	3	60	40	100
PEC	BTECCE23605B	Internet of Things	3	0	0	3	60	40	100
PEC	BTECCE23606A	Machine Learning Lab	0	0	2	1	15	10	25
PEC	BTECCE23606B	Internet of Things Lab	0	0	2	1	15	10	25
PEC	BTECCE23607A	Ethical Hacking and Penetration Testing	3	0	0	3	60	40	100
PEC	BTECCE23607B	Real Time Operating System	3	0	0	3	60	40	100
PEC	BTECCE23608A	Ethical Hacking and Penetration Testing Lab	0	0	2	1	15	10	25
PEC	BTECCE23608B	Real Time Operating System Lab	0	0	2	1	15	10	25
VSEC	BTECCE23609	Advance Web and Server Technology	1	0	2	2	50	0	50
MC	MT23CE04A	Cloud Computing	3	0	0	3	60	40	100
TOTAL			16	0	10	0	410	240	650
			21						
Instructions, if any: 1 Theory/Tutorial Hour = 1 Credit, 2 Practical hours = 1 Credit									

Hons	BTECCE23H610	Cloud and Mobile Security	3	0	0	3	60	40	100
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Name of Head of Department

Sign:

Date:

Name of Dean

Sign:

Date:

SEM V

BTECCE23501:: Operating Systems

Course Type	PCC	Semester	5
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Teaching Scheme		Credits		Examination Scheme	
Lecture	3Hr./Week	Lecture	3	CIE Marks	60
Tutorial	--Hr./Week	Tutorial	-	ESE Marks	40
Practical/Studio	--Hr./Week	Practical/Studio	-	Total Marks	100
Total	3Hr./Week	Total	3		

Course Description

This course helps the student to identify the role of operating systems and explain the different structures of operating systems. It highlights the OS support for processes and threads, depicts scheduling algorithms and deadlock mechanisms. It also identifies virtual memory requirement of an operating system and illustrates the importance of I/O and file systems.

Course Outcomes

CO No.	Statement
1	Examine the functions of a contemporary Operating system with respect to convenience, efficiency and the ability to evolve.
2	Demonstrate knowledge in applying system software and tools available in modern operating system, such as threads, system calls, semaphores for software development.
3	Apply various CPU scheduling algorithms and identify deadlock mechanisms to construct solutions to real world problems.
4	Understand the organization of memory and memory management hardware.
5	Analyze I/O and file management techniques for better utilization of secondary memory.

Mapping of COs to POs and PSOs

CO No	POs												PSOs				BTL
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4	
CO1	2																UN
CO2		2															AN
CO3						3											AP
CO4							3										AN
CO5															3		AP

Affinity Level: 1- Slight, 2- Moderate, 3-Substantial, BTL: Bloom's Taxonomy Level

Course Content

Unit No 1	Introduction	Hours 9	CO	BTL
	Introduction to OS: What is OS, Interaction of OS and hardware, Goals of OS, Basic functions of OS, OS Services, System Calls. Types of OS: Batch, Multiprogramming, Time sharing, Parallel, Distributed & Real-time OS. Real Time OS: Concepts, Design Issues, Architecture, Real Time Kernel Embedded Systems: Definition, Classification, Characteristics, Components: Microcontrollers, sensors, actuators, communication modules, and memory. Embedded System Design cycle. Virtualization technologies: Hypervisor Containerization platforms: Docker, Kubernetes Shell: Linux commands and shells, shell programming, AWK programming. Introduction to Mobile OS: Architecture & Overview of Android OS.		1	UN
Unit No 2	Process Management	Hours 9	2	AN
	Processes: Process Concept, Process State Model, Process Description Threads: Multithreading models, Symmetric Multiprocessing, User level and kernel level threads, Microkernels. Concurrency: Issues with concurrency, Principles of Concurrency Mutual Exclusion: H/W approaches, S/W approach, OS/Programming Language support: Semaphores, Mutex, Monitors. Classical Problems of Synchronization: Readers-Writers problem, Producer Consumer problem, Dining Philosopher problem.			
Unit No 3	CPU Scheduling and Deadlock	Hours 9	3	AP
	Uniprocessor Scheduling: Types of Scheduling - Preemptive, Non-preemptive, Long-term, Medium-term, Short-term, Scheduling Criteria. Scheduling Algorithms: FCFS, SJF, RR, Virtual Round Robin, Priority Real time Scheduling Algorithms: Earliest Deadline First, Rate Monotonic Scheduling, Least Laxity First Deadlock: Principles of deadlock, Deadlock Prevention, Deadlock Avoidance, Deadlock Detection, Deadlock Recovery. Case studies: Scheduling in LINUX and Windows Case studies: Deadlock management in LINUX and Windows			
Unit No 4	Memory Management	Hours 9	4	AN
	Memory Management concepts: Memory Management requirements Memory Partitioning: Fixed, Dynamic Partitioning, Buddy Systems. Placement Strategies: First Fit, Best Fit, and Worst Fit. Fragmentation, Swapping, Thrashing. Virtual Memory: Concepts, Address translation, VM with Paging, Page Table Structure, Translation Look-aside Buffer, VM with Segmentation, Working Set Page Replacement Policies: FIFO, LRU, Optimal.			

Case studies: Memory management implementation in LINUX and Windows				
Unit No 5	I/O & Device Management	Hours 9	5	AP
I/O management: I/O Devices - Types, Characteristics of Serial and Parallel devices, OS design issues for I/O management, I/O Buffering.				
Disk Scheduling: FCFS, SCAN, C-SCAN, SSTF.				
File Management: Concepts, File Organization, File Directories, File Sharing. Record Blocking, Secondary Storage Management, Free Space management, Security.				
File System: Structure, Implementation, Memory mapped files, Study of LINUX File system				

Textbooks

1	<i>“Operating Systems”, Stallings William, Pearson Education, ISBN: 0-13-031999-6, 5th Edition.</i>
2	<i>“Operating System Concepts”, Silberschatz A., Galvin P., Gagne G., John Wiley and Sons, ISBN: 9971-51-388-9, 6th Edition.</i>

Reference Books / Journal Articles / Weblink

1	<i>“Modern Operating Systems”, Tanenbaum Andrew S., Prentice Hall India, ISBN: 81-7808-447-3, 2nd Edition.</i>
2	<i>“Unix Concepts and Applications”, Das Sumitabha, Tata McGraw Hill, ISBN: 0-07-053475-6, 3rd Edition.</i>
3	<i>“Unix Shell Programming”, Yashavant Kanetkar, BPB Publications, ISBN: 81-7029-753-2, 1st Edition.</i>
4	<i>"Embedded systems: Architecture, programming and design", Rajkamal, McGraw-Hill, ISBN: 978-0070588440, 2003.</i>
5	<i>“Real-time systems”, Jane W. S. Liu, Prentice Hall, 2000.</i>

BTECCE23502:: Operating Systems Lab

Teaching Scheme		Credits		Examination Scheme	
Lecture	-- Hr./Week	Lecture	-	CIE Marks	15
Tutorial	--Hr./Week	Tutorial	-	ESE Marks	10
Practical/Studio	2 Hr./Week	Practical/Studio	1	Total Marks	25
Total	2 Hr./Week	Total	1		

List of Experiments

1	Execution of basic and advanced Unix commands (Hackerrank challenges - Linux shell: Bash, Text processing)
2	Implement following shell programs a. To find out if a given string is a palindrome or not. b. To sort numbers of a given array using bubble sort
3	Generate a student report using Awk programming (Hackerrank challenges - Linux shell: Grep Sed Awk)
4	Solve the Readers-Writers problem a. using threads and semaphores b. using threads and mutex
5	Solve the Producers-Consumers problem a. using threads and semaphores b. using threads and mutex
6	Implement the Dining Philosopher's problem using Multithreading
7	Simulate the following CPU scheduling algorithms: a. First come First serve b. Shortest Job First (Non-preemptive) c. Shortest Job First (Preemptive) d. Round Robin e. Priority (Non-preemptive) f. Priority (Non-preemptive) (Draw the Gantt charts and display the finish time, turnaround time, waiting time for each process)
8	Simulate the following Real time Scheduling Algorithms: a. Earliest Deadline First b. Rate Monotonic Scheduling c. Least Laxity First
9	Implement Banker's Safety algorithm for Deadlock Avoidance
10	Simulate the following page replacement algorithms a. FIFO b. LRU c. OPT

BTECCE23503:: Computer Network

Course Type	PCC	Semester	5
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Teaching Scheme		Credits		Examination Scheme	
Lecture	3 Hr./Week	Lecture	3	CIE Marks	60
Tutorial	-- Hr./Week	Tutorial	-	ESE Marks	40
Practical/Studio	0 Hr./Week	Practical/Studio	0	Total Marks	100
Total	3 Hr./Week	Total	3		

Course Description

This course helps the students to design, analyze, and implement computer networks across multiple layers, covering network fundamentals, IPv4/IPv6, TCP/UDP, DNS/SMTP/HTTP, and wireless networks like WLANs, Bluetooth, and WiMax.

Course Outcomes

CO No.	Statement
1	students will be able to design, analyze, and implement computer networks by understanding network fundamentals, protocol suites, and media access control protocols.
2	students will be able to design and implement network layer protocols and routing algorithms, including IPv4, IPv6, subnetting, and various routing protocols.
3	students will be able to analyze and implement transport layer protocols, including TCP and UDP, and apply socket programming principles to ensure reliable data transfer and quality of service.
4	students will be able to design and utilize application layer protocols such as DNS, SMTP, HTTP, and others to facilitate network communication and management.
5	students will be able to design and analyze wireless networks, including WLANs, Bluetooth, WiMax, and wireless sensor networks, understanding their architectures and applications.

Mapping of COs to POs and PSOs

CO No	POs												PSOs				BT L
	P O 1	P O 2	P O 3	P O 4	P O 5	P O 6	P O 7	P O 8	P O 9	P O 10	P O 11	P O 12	P S O 1	P S O 2	P S O 3	P S O 4	
CO1	3																UN
CO2		2															AP
CO3					2												AN
CO4			3														AP
CO5																3	UN

Affinity Level: 1- Slight, 2- Moderate, 3-Substantial, BTL: Bloom's Taxonomy Level

Course Content

Unit No 1	Introduction to Physical Layer and Data Link Layer	Hours 9	CO	BTL
	Types of Computer Network, Network Topologies, Overview of protocol suite: OSI and TCP/IP, infrastructure network, ad-hoc network. Physical layer: Functions, Transmission media, Twisted Pair, Coaxial and Fiber-optic cables, Ethernet. Data Link Layer: Functions, Design Issues, Error Detection and correction, Examples on Checksum, Stop-and-Wait protocol, Sliding Window protocols, HDLC. Medium access Control :ALOHA, CSMA, Ethernet		1	UN
Unit No 2	Network Layer	Hours 9	2	AP
	Functions, Design Issues, Packet switching, Connectionless and Connection-oriented Services, Virtual Circuit and Datagram Subnets. Autonomous system. IP Protocols: IPv4, IPv6, Subnetting, CIDR. Routing Algorithms: Optimality principle, Static routing, Dynamic routing, Distance Vector routing, link state routing, path vector routing. Protocols: ARP, RARP, ICMP, IGMP Unicast Routing Algorithms: RIP, OSPF, BGP, EIGRP, MPLS.			
Unit No 3	Transport Layer	Hours 9	3	AN
	Transport layer: Transport layer services (Duties), TCP: TCP header, Services, Segments, Connection Establishment, Flow control, Congestion Control, Congestion Control Algorithms, Leaky Bucket, Token Bucket and QoS, Timers, UDP: UDP header, Datagram, Services, Applications, Socket: Primitives, TCP & UDP Sockets.			
Unit No 4	Application Layer	Hours9	4	AP
	Domain Name System (DNS) and DNS servers, MIME, SMTP, Mail Gateways, Remote login, File Transfer Protocol, SNMP, DHCP, HTTP, Telnet.			
Unit No 5	Wireless Network	Hours 9	5	UN
	Wireless LANs: Fundamentals of WLAN, Design goals, Characteristics, Network Architecture, IEEE 802.11: components in IEEE 802.11 network, Bluetooth: Architecture Layers, operational states, IEEE 802.16 WiMax: Services, Architecture, Layers, comparison between Bluetooth, IEEE 802.11 and IEEE 802.16 Wireless Sensor Network: Applications of Sensor Network, Sensor node architecture Issues and Challenges in Designing a Sensor Network			

Textbooks

1	Andrew S. Tanenbaum, "Computer Networks", 4th Edition, PHI, ISBN 81-203-2175-8.
2	Forouzan B, "Data communication and Computer Networks", Tata McGraw Hill

Reference Books / Journal Articles / Weblink

1	Kurose, Ross, "Computer Networking-a top down approach featuring the internet ", Pearson Education. ISBN: 978-81-7758-878
2	Comer D, "Computer networks and internet", Pearson Education

3	<i>Behrouz A. Forouzan, TCP/IP Protocol Suite, McGraw Hill Education, ISBN: 978-0-07-070652-1, 4th Edition.</i>
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BTECCE23504: Computer Network Laboratory Lab

Course Type	PCC	Semester	5
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Teaching Scheme		Credits		Examination Scheme	
Lecture	- Hr./Week	Lecture	-	CIE Marks	15
Tutorial	-- Hr./Week	Tutorial	-	ESE Marks	10
Practical/Studio	2 Hr./Week	Practical/Studio	1	Total Marks	25
Total	2 Hr./Week	Total	1		

List of Experiments

1.	Set up a small network of 2 to 4 computers using Switch. It includes installation of LAN Cards, Preparation of Cables, Assigning IP addresses and sharing C drive.
2.	Studying Linux and Windows network commands. [ping, pathping, ipconfig/ifconfig, arp, netstat, nbtstat, nslookup, route, traceroute/tracert, nmap, etc]
3.	Using a Network Simulator (e.g. packet tracer) Configure subnetting and supernetting
4.	Configuration of router by using router commands and implement static routing
5.	Configuration of router for implementation of RIP protocol
6.	Configuration of router for implementation of OSPF protocol
7.	Configuration of router for implementation of EIGRP protocol
8.	Configuration of Wireless access point/Router with static IP addressing and DHCP with MAC security and filters.
9.	Installing and configuring DHCP server for Linux/Windows.
10.	Socket Programming for TCP Client and TCP Server.
11.	Socket Programming for UDP Client, UDP Server.

BTECCE23505::Design and Analysis of Algorithms

Course Type	PCC	Semester	5
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Teaching Scheme		Credits		Examination Scheme	
Lecture	2 Hr./Week	Lecture	2	CIE Marks	50
Tutorial	-- Hr./Week	Tutorial	-	ESE Marks	0
Practical/Studio	--Hr./Week	Practical/Studio	-	Total Marks	50
Total	2Hr./Week	Total	2		

Course Description

This course will help students to understand and apply the algorithm analysis techniques and efficiency of alternative algorithmic solutions for the same problem. It will also identify different algorithm design techniques and algorithm analysis techniques.

Course Outcomes

CO No.	Statement
1	Analyze the performance of algorithms.
2	Choose appropriate algorithm design techniques for solving problems.
3	Understand how the choice of data structures and the algorithm design methods impact the performance of programs along with the principle of optimality.
4	Illustrate various problems using backtracking.
5	Compare different methods of branch and bound strategy.

Mapping of COs to POs and PSOs

CO No	POs												PSOs				BTL
	P O 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3	PSO 4	
CO1	3																AP
CO2		3															AP
CO3					2												AN
CO4						3											AN
CO5															3		AN

Affinity Level: 1- Slight, 2- Moderate, 3-Substantial, BTL: Bloom's Taxonomy Level

Course Content

Unit No 1	Basic introduction, time complexity analysis	Hours 8	CO	BTL
	Basic introduction, time complexity analysis, Asymptotic notations (Big Oh, small oh, Big Omega, Theta notations). Best case, average case, and worst case time and space complexity of		1	AP

algorithms. Overview of searching, sorting algorithms. Adversary lower bounds (for comparison based sorting). Recurrence relations: Substitution method, Recursion Tree method, Master's theorem and applications. Mathematical Induction to get asymptotic bounds on time complexity. Real World Case Studies by Hackerrank: Searching, Recursion.				
Unit No 2	Divide and Conquer	Hours 6	2	AP
Analyzing Quick sort, Randomized Quick sort, merge sort, Counting Inversions, finding majority element, Finding Median, Computing convex hull of points in plane, Graham's scan algorithm. Real World Case Studies by Hackerrank: Sorting.				
Unit No 3	Greedy Technique and Dynamic Programming	Hours 8	3	AP
Greedy method: General method, Applications: Fractional Knapsack problem, 0/1 Knapsack problem, Minimum cost spanning trees: Prim's algorithm and Kruskal's Algorithm, Single source shortest path problem: Dijkstra's algorithm, Optimal Merge pattern, Huffman Trees. Dynamic programming: Principle of optimality, Multi stage graph, Optimal Binary Search Trees, Knapsack Problem. Real World Case Studies by Hackerrank: Greedy and Dynamic Programming.				
Unit No 4	Backtracking Strategy	Hours 4	4	AN
Backtracking: General strategy, N-queens problem, graph coloring, subset sum problem, Knapsack problem, Hamiltonian cycle. Real World Case Studies by Hackerrank: Graph Theory, Game Theory.				
Unit No 5	Branch and Bound	Hours 4	5	AN
Branch and Bound: General Method, applications - 0/1 Knapsack problem, LC Branch and Bound solution, FIFO Branch and Bound solution, Travelling sales person problem. NP-Hard and NP-Complete problems Real World Case Studies by Hackerrank: NP Complete				

Textbooks

1	<i>Introduction to Algorithms" by Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, and Clifford Stein</i>
2	<i>Algorithm Design" by Jon Kleinberg and Éva Tardos</i>

Reference Books / Journal Articles / Weblink

1	<i>"The Algorithm Design Manual" by Steven S. Skiena</i>
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BTECCE23506A : Compiler Design

Course Type	PEC	Semester	6
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Teaching Scheme		Credits		Examination Scheme	
Lecture	3 Hr./Week	Lecture	3	CIE Marks	60
Tutorial	-- Hr./Week	Tutorial	-	ESE Marks	40
Practical/Studio	0 Hr./Week	Practical/Studio	0	Total Marks	100
Total	3 Hr./Week	Total	3		

Course Description

This course helps to students to study the functionalities and the design of Pass structure of compiler, understanding of code Optimization and parallel compilation techniques

Course Outcomes

CO No.	Statement
1	Identify and Interpret the different phases of a compiler and their functioning.
2	Design a well-structured system to ensure the syntactic correctness of a program.
3	Deploy efficient techniques for semantic analysis to generate intermediate code.
4	Apply code optimization transformations to improve the performance of target code
5	Propose techniques to generate machine code which conforms to the target machine specifications.

Mapping of COs to POs and PSOs

CO No	POs												PSOs				BT L
	P O 1	P O 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3	PSO 4	
CO1	2																UN
CO2			2														AP
CO3						2											AN
CO4				2													AP
CO5															2		AN

Affinity Level: 1- Slight, 2- Moderate, 3-Substantial, BTL: Bloom's Taxonomy Level

Course Content

Unit No 1	Introduction	Hours7	CO	BTL
Introduction to Compiler, Interpreters, Phases and Passes, Bootstrapping, Cross Compiler Lexical Analyzer: Specification and Recognition of Tokens, Regular expressions, NFA Vs DFA, Look ahead operation, Lexical Phase errors, , Automatic construction of lexical analyzer using LEX, LEX features and specification.			1	UN
Unit No 2	Syntax and Semantic Analysis	Hours 9	2	AP
Expressing Syntax: CFG, Top-Down Parsing: Recursive Descent, Predictive Parsers. Bottom-Up Parsing: LR Parsers: Constructing SLR parsing tables, Constructing Canonical LR parsing tables, Constructing LALR parsing tables, Using ambiguous grammars. Parser generator YACC, Symbol-Table Structure, Error Detection & Recovery strategies.				
Unit No 3	Syntax-Directed Translation and Intermediate Code Generation	Hours 10	3	AN
Syntax-Directed Definitions , Attribute grammar, Dependency graphs, S and L attributed grammar, Bottom-Up Evaluation, Top-Down Translation Intermediate Representations – Need, Forms, SDT scheme for Intermediate Code Generation of assignment statement, declaration statement. Control flow translation of Boolean expression (Back patching), if, if-else statement, while statement, array assignment. Error Detection & Recovery: semantic errors.				
Unit No 4	Code Optimization	Hours 9	4	AP
Introduction: Principal Sources of Optimization, Optimization of basic Blocks (local and global) Data Flow Analysis: data flow equations, reaching definitions, live variable analysis, available expressions, Loops in flow graphs: Dominators, Back edges and Reducibility.				
Unit No 5	Code Generation and Introduction to advanced topics	Hours 10	5	AN
Issues in Code Generation, Target Language, Basic Blocks and Flow Graphs, Next-use information, Simple Code generator, DAG representation of Basic Blocks, Register allocation and Assignment, Peephole Optimization. Automatic Parallelization, Instruction Scheduling, Software Pipelining. Introduction to advanced topics – Just-In-Time Compilation, Dynamic compilation, Interpreters (JVM/ Dalvik), Cross compilation using XMLVM, Case studies (Self Study): GCC, g++, LLVM.				

Textbooks

1	<i>A. V. Aho, M. S. Lam, R. Sethi, J. D. Ullman; “Compilers: Principles, Techniques and</i>
2	<i>K. Cooper, L. Torczon, Morgan Kaufmann; “Engineering a Compiler”, ISBN 1-55860-698-X, First Edition, 2003.</i>

Reference Books / Journal Articles / Weblink

1	<i>S. S. Muchnik, Morgan Kaufmann; “Advanced Compiler Design and Implementation”, ISBN 8178672413, First Edition, 1997.</i>
2	<i>K. Muneeswaran : “ Compiler Design”, ISBN 978-0-19-806664-4, Oxford University Press,2013</i>
3	<i>J. R. Levine, T. Mason, D. Brown; “Lex & Yacc”, O’Reilly, ISBN1-56592-000-7, Second Edition, 1992.</i>

BTECCE23507A: Compiler Design Lab

Teaching Scheme		Credits		Examination Scheme	
Lecture	0Hr./Week	Lecture	0	CIE Marks	15
Tutorial	--Hr./Week	Tutorial	-	ESE Marks	10
Practical/Studio	2Hr./Week	Practical/Studio	1	Total Marks	25
Total	2Hr./Week	Total	1		

List of Experiments

1	Assignment to understand basic syntax of LEX specifications, built-in functions and Variables.
2	Implement a Lexical Analyzer using LEX for the subset of C Language.
3	Implement a parser for an expression grammar using YACC and LEX. Extend to handle variables with single letter names.
4	Generate and populate appropriate Symbol Table.
5	YACC specification to check the syntax of language statements with error detection.
6	Implementation of Semantic Analysis Operations (like type checking, verification of function parameters, variable declarations and coercions) possibly using an Attributed Translation Grammar.
7	Implement the front end of a compiler that generates the three address code for a simple language.
8	Implement Common Sub-expression Elimination and Copy Propagation optimizations for an input 3-address code.
9	Implement Dead-Code Elimination, Loop optimizations for input 3-address code
10	Generate an appropriate Target Code from the given intermediate code assuming suitable processor details.

BTECCCE23506B:: Embedded System Design

Course Type	PEC	Semester	6
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Teaching Scheme		Credits		Examination Scheme	
Lecture	3Hr./Week	Lecture	3	CIE Marks	60
Tutorial	--Hr./Week	Tutorial	-	ESE Marks	40
Practical/Studio	0Hr./Week	Practical/Studio	0	Total Marks	100
Total	3Hr./Week	Total	3		

Course Description

This course provides an in-depth introduction to the design and development of embedded systems, focusing on both hardware and software components. Embedded systems are specialized computing systems that are designed to perform specific tasks within a larger system. The course covers the fundamental concepts of embedded systems, including microcontrollers, real-time operating systems, sensors, actuators, and communication protocols.

Students will learn how to design embedded systems from the ground up, starting with hardware selection and integration, and progressing to writing efficient embedded software. The course will also emphasize debugging, testing, and optimization techniques to ensure reliable system performance in resource-constrained environments.

Course Outcomes

CO No.	Statement
1	Understanding the basic principles of embedded systems and their design.
2	Designing basic microcontroller-based programs for various I/O operations.
3	Proficiency in C programming tailored for embedded systems.
4	Simulating communication between devices using software tools.
5	Proficiency in using simulation tools for design, testing, and optimization.

Mapping of COs to POs and PSOs

CO No	POs												PSOs				BT L
	P O 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PS O1	PSO 2	PSO 3	PSO 4	
CO1	3																UN
CO2		2															AP
CO3			2														AN
CO4					3												AP
CO5															3		AN

Affinity Level: 1- Slight, 2- Moderate, 3-Substantial, BTL: Bloom's Taxonomy Level

Course Content

Unit No 1	Introduction to Embedded Systems	Hours 9	CO	BTL
	Definition of Embedded System, Embedded Systems Vs General Computing Systems, History of Embedded Systems, Classification, Major Application Areas, Purpose of Embedded Systems, Characteristics and Quality Attributes of Embedded Systems. Applications of Embedded Systems in various industries (automotive, medical, consumer electronics, industrial automation), Components of Embedded Systems: Microcontrollers, sensors, actuators, communication modules, and memory, Embedded System Design Cycle: Problem definition, design, testing, and deployment, Introduction to Embedded System Software: Basic principles of embedded system programming (C and assembly), Real-Time Constraints: Real-time systems, task scheduling, and resource management.		1	UN
Unit No 2	Microcontroller Architecture and Programming	Hours 9	2	AP
	Microcontroller Architecture: Overview of popular microcontrollers (e.g., 8051, ARM, AVR), Registers, Memory, and I/O Operations: Internal memory (RAM, ROM), GPIO, timers, interrupts, Programming Microcontrollers: Introduction to microcontroller programming using C, Interrupts and Timers: Types of interrupts, interrupt service routines, and timer-based applications, Analog-to-Digital and Digital-to-Analog Conversion: Basic concepts of ADC/DAC and their software implementation, Basic I/O Interfacing: LED blink, switches, and basic input/output operations using software.			
Unit No 3	Embedded C Programming and Real-Time Operating Systems (RTOS)	Hours 9	3	AN
	C Programming for Embedded Systems: Functions, arrays, structures, pointers, and memory management specific to embedded systems, Real-Time Operating Systems (RTOS): Concepts of multitasking, scheduling, and context switching in embedded systems, Task Management in RTOS: Task creation, priority-based scheduling, and inter-task communication, RTOS Frameworks: Overview of popular RTOS platforms (e.g., FreeRTOS), RTOS without Hardware Support: Simulating RTOS in software, task scheduling using software emulators.			
Unit No 4	Embedded System Communication Protocols	Hours 9	4	AP
	Introduction to Communication Protocols: Overview of serial communication, I2C, SPI, UART, Simulation of Communication Protocols: Using software to simulate UART, I2C, and SPI communication, Interfacing with Sensors and Actuators: Software simulation of interfacing with sensors (temperature, pressure, etc.) and actuators (motors, LEDs), Data Exchange and Buffering: Implementing software-based protocols for data exchange and buffering, Error Detection and Handling in Communication: Software strategies to handle errors in communication protocols.			
Unit No 5	Design and Testing of Embedded Systems	Hours 9	5	AN
	Design Methodologies: Top-down vs. bottom-up approaches in embedded system design, Software Testing and Debugging: Techniques for testing embedded system code, including unit testing, integration testing, and debugging using software tools (e.g., debuggers, simulators), Optimizing Embedded System Code: Memory and time optimization in embedded systems, power efficiency through software algorithms, Case Studies: Discuss real-world embedded systems case studies, including automotive, medical, and industrial			

control systems, Simulation Tools: Overview of software tools used for embedded system design (e.g., Proteus, MPLAB X IDE, Keil, and Simulink).		
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Textbooks

1	Valvano, J. W. (2014). <i>Embedded systems: Introduction to ARM Cortex-M microcontrollers</i> . Addison-Wesley. ISBN: 978-1-450-14216-6
2	Valvano, J. W. (2015). <i>Embedded systems: Real-time operating systems for ARM Cortex-M microcontrollers</i> . Addison-Wesley. ISBN: 978-0-13-439064-5
3	Marwedel, P. (2011). <i>Embedded systems design</i> . Springer. ISBN: 978-1-4471-2212-5
4	Ganssle, J. (2011). <i>The art of designing embedded systems</i> . Newnes. ISBN: 978-0-7506-7031-5
5	Simon, D. E. (2001). <i>C programming for embedded systems</i> . Elsevier. ISBN: 978-0750672929

Reference Books

1	Rajkamal. (2003). <i>Embedded systems: Architecture, programming and design</i> . McGraw-Hill. ISBN: 978-0070588440
2	Barr, M., & Massa, A. (2006). <i>Programming embedded systems in C and C++</i> . O'Reilly Media. ISBN: 978-0596009835
3	Peckol, J. K. (2008). <i>Embedded systems: A contemporary design tool</i> . Wiley. ISBN: 978-0471734600
4	Laplane, P. A. (2009). <i>Real-time systems design and analysis</i> . Wiley. ISBN: 978-0763748885
5	Yiu, J. (2014). <i>The definitive guide to ARM Cortex-M3 and Cortex-M4 processors</i> . Elsevier. ISBN: 978-0124080829
6	Valvano, J. W. (2017). <i>Designing embedded systems and the Internet of Things (IoT) with the ARM mbed</i> . Addison-Wesley. ISBN: 978-0134423775

BTECCCE23507B:: Embedded System Design Lab

Teaching Scheme		Credits		Examination Scheme	
Lecture	0Hr./Week	Lecture	0	CIE Marks	15
Tutorial	--Hr./Week	Tutorial	-	ESE Marks	10
Practical/Studio	2Hr./Week	Practical/Studio	1	Total Marks	25
Total	2Hr./Week	Total	1		

List of Experiments

1	Simulate the behavior of an LED turning ON and OFF in a loop with a delay using Python.
2	Create a Python program that mimics an interrupt by calling a function every 2 seconds (simulate a timer interrupt). The function can simply print a message like "Timer Interrupt Triggered".
3	Create a Python program where the user simulates a button press (e.g., by entering a specific key or number). The program should toggle the LED (simulated by printing a message) when the button is pressed.
4	Write a Python program that maps an analog value (between 0-1023) to a corresponding digital value using a simple conversion formula (e.g., dividing by 1023). Simulate a DAC by converting a digital value back to an analog one.
5	Write a Python program to simulate task scheduling by creating multiple functions (tasks) that run in sequence with specified time intervals (using time.sleep()). Simulate different priority tasks using simple priority control.
6	Simulate UART communication by writing a Python program to send and receive strings between two simulated devices. Use Python's socket or queue libraries to simulate the sending and receiving of messages.
7	Write a Python program to simulate I2C communication where one device (master) sends data to another device (slave) via I2C protocol. Simulate the data transfer and acknowledge signals using Python data structures.
8	Create a Python program to simulate SPI communication, where a master sends a byte of data to a slave, and the slave sends an acknowledgment message. Use Python's list or queue to simulate the data exchange.
9	Write a Python program that continuously displays the current time (simulating a real-time clock) and adjusts the time by 1 second each time. Use time.sleep() to create a time delay.
10	Simulate a communication protocol where data is sent with possible errors. Use checksums or parity bits to detect errors in the received data. If an error is detected, handle it and request retransmission.

MT23CE03A:: Software Engineering

Course Type	MC	Semester	5
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Teaching Scheme		Credits		Examination Scheme	
Lecture	3 Hr./Week	Lecture	3	CIE Marks	60
Tutorial	0Hr./Week	Tutorial	0	ESE Marks	40
Practical/Studio	-- Hr./Week	Practical/Studio	-	Total Marks	100
Total	3 Hr./Week	Total	3		

Course Description

This course helps to students to analyze the target system properties and recommend solution alternative by practicing project planning, scheduling, estimation, risk management activities and software testing

Course Outcomes

CO No.	Statement
1	CO1: Apply software engineering principles to develop software.
2	CO2: Analyze software requirements and formulate design solution for a software
3	CO3: Explain concepts of project estimation, planning and scheduling.
4	CO4: Explain risk management and software configuration management.
5	CO5: Explain various types of software testing.

Mapping of COs to POs and PSOs

CO No	POs												PSOs				BTL
	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PS O1	PS O2	PS O3	PS O4	
CO1			2														UN
CO2									3								AP
CO3										2							AP
CO4											3						AP
CO5															3		AP

Affinity Level: 1- Slight, 2- Moderate, 3-Substantial, BTL: Bloom's Taxonomy Level

Course Content

Unit No 1	Introduction to Software Engineering and Software Process Models	Hours 9	CO	BTL
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Software Engineering Fundamentals: Introduction to software engineering, The Nature of Software, Defining Software, Software Engineering Practice. Software Process: A Generic Process Model, defining a Framework Activity, Identifying a Task Set, Process Patterns, Process Assessment and Improvement, Prescriptive Process Models, The Waterfall Model, Incremental Process Models, Evolutionary Process Models, Concurrent Models, A Final Word on Evolutionary Processes. Unified Process, Agile software development: Agile methods, plan driven and agile development. #Exemplar/Case Studies- Agile Tools- JIRA			1	UN
Unit No 2	Software Requirements Engineering and Analysis	Hours 9	2	AP
Modelling: Requirements Engineering, Establishing the Groundwork, Identifying Stakeholders, Recognizing Multiple Viewpoints, working toward Collaboration, Asking the First Questions, Eliciting Requirements, Collaborative Requirements Gathering, Usage Scenarios, Elicitation Work Products, Developing Use Cases, Building the Requirements Model, Elements of the Requirements Model, Negotiating Requirements, Validating Requirements. Suggested Free Open Source tools: StarUML, Modelio, SmartDraw. #Exemplar/Case Studies- Study SRS of Online Voting system (http://dos.iitm.ac.in/OOSD_Material/CaseStudies/CaseStudy2/eVote-srs.pdf), Library management System				
Unit No 3	Estimation and Scheduling	Hours 9	3	AP
Estimation for Software Projects: The Project Planning Process, Defining Software Scope and Checking Feasibility, Resources management, Reusable Software Resources, Environmental Resources, Software Project Estimation, Decomposition Techniques, Software Sizing, Problem-Based Estimation, LOC-Based Estimation, FP-Based Estimation, Object Point (OP)-based estimation, Process-Based Estimation, Process-Based Estimation, Estimation with Use Cases, Use-Case-Based Estimation, Reconciling Estimates, Empirical Estimation Models, The Structure of Estimation Models, The COCOMO II Mode, Preparing Requirement Traceability Matrix Project Scheduling: Project Scheduling, Defining a Task for the Software Project, Scheduling. Suggested Free Open Source Tool: GanttProject, Agantty, ProjectLibre. #Exemplar/Case Studies- Study SRS of Online Voting system (http://dos.iitm.ac.in/OOSD_Material/CaseStudies/CaseStudy2/eVote-srs.pdf), Library management System,				
Unit No 4	Design Engineering	Hours 9	4	AP
Design Concepts: Design within the Context of Software Engineering, The Design Process, Software Quality Guidelines and Attributes, Design Concepts - Abstraction, Architecture, design Patterns, Separation of Concerns, Modularity, Information Hiding, Functional Independence, Refinement, Aspects, Refactoring, Object-Oriented Design Concept, Design Classes, The Design Model, Data Design Elements, Architectural Design Elements, Interface Design Elements, Component-Level Design Elements, Component Level Design for WebApps, Content Design at the Component Level, Functional Design at the Component Level, Deployment-Level Design Elements, Architectural Design: Software Architecture, What is Architecture, Why is Architecture Important, Architectural Styles, A brief Taxonomy of Architectural Styles. Suggested Free Open Source Tool: SmartDraw. #Exemplar/Case Studies - Study design of Biometric Authentication software				
Unit No 5	Software Testing	Hours 9	5	AN
A Strategic Approach to Software Testing, Verification and Validation, Organizing for Software Testing, Software Testing Strategy—The Big Picture, Criteria for Completion of Testing, Strategic Issues, Test Strategies for Conventional Software, Unit Testing, Integration				

Testing, Test Strategies for Object-Oriented Software, Unit Testing in the OO Context, Integration Testing in the OO Context, Test Strategies for WebApps, Validation Testing, Validation-Test Criteria, Configuration Review. Suggested Free Open Source Tools: Selenium, JUnit. #Exemplar/Case Studies - Selenium Testing with any online application		
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Textbooks

1	<i>Ian Sommerville, "Software Engineering", Addison-Wesley, 9th Edition, 2010, ISBN-13: 978-0137035151.</i>
2	<i>Roger S Pressman, "Software Engineering: A Practitioner's Approach", McGraw Hill, 6/e, 2005</i>

Reference Books / Journal Articles / Weblink

1	<i>Carlo Ghezzi, —Fundamentals of Software Engineering", Prentice Hall India, ISBN-10: 0133056996</i>
2	<i>Rajib Mall, —Fundamentals of Software Engineering", Prentice Hall India, ISBN-13: 978 8120348981</i>
3	<i>Pankaj Jalote, —An Integrated Approach to Software Engineering", Springer, ISBN 13: 9788173192715.</i>

OE23CE03:: Software Project Management

Course Type	OE		Semester	4
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Teaching Scheme		Credits		Examination Scheme	
Lecture	3Hr./Week	Lecture	3	CIE Marks	60
Tutorial	1Hr./Week	Tutorial	1	ESE Marks	40
Practical/Studio	0Hr./Week	Practical/Studio	0	Total Marks	100
Total	4Hr./Week	Total	4		

Course Description

This course is aimed at introducing the primary important concepts of project management related to managing software development projects. Students will also get familiar with the different activities involved in Software Project Management. Further, they will also come to know how to successfully plan and implement a software project management activity, and to complete a specific project in time with the available budget.

Course Outcomes

CO No.	Statement
1	To define the principles of project management for developing software.
2	To explain various project management scheduling techniques.
3	To apply different techniques of project monitoring, control and review.
4	To classify various project management tools and estimate the risks involved in project activities.
5	To Assess issues related to project quality and staffing.
6	To discuss the effect of project management practices in an organization

Mapping of COs to POs and PSOs

CO No	POs												PSOs				BTL
	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3	PSO 4	
CO1	3																UN
CO2		2															AP
CO3				3													AN
CO4					3												AP
CO5															3		AN

Affinity Level: 1- Slight, 2- Moderate, 3-Substantial, BTL: Bloom's Taxonomy Level

Course Content

Unit No 1	Introduction to Software Project Planning	Hours 9	CO	BTL
	Fundamentals of Software Project Management (SPM), Need Identification, Vision and Scope Document, Project Management Cycle, SPM Objectives, SPM Framework, Software Project Planning, Planning Objectives, Project Plan, Types of Project Plan, Structure of a Software Project Management Plan, Software Project Estimation, Estimation Methods, Estimation Models, Decision Process		1	UN
Unit No 2	Project Organization and Scheduling Project Elements	Hours 9	2	AP
	Work Breakdown Structure (WBS), Types of WBS, Functions, Activities and Tasks, Project Life Cycle and Product Life Cycle, Ways to Organize Personnel, Project Schedule, Scheduling Objectives, Building the Project Schedule, Scheduling Terminology and Techniques, Network Diagrams: PERT, CPM, Bar Charts: Milestone Charts, Gantt Charts			
Unit No 3	Project Monitoring and Control	Hours 9	3	AN
	Dimensions of Project Monitoring & Control, Earned Value Analysis, Earned Value Indicators: Budgeted Cost for Work Scheduled (BCWS), Cost Variance (CV), Schedule Variance (SV), Cost Performance Index (CPI), Schedule Performance Index (SPI), Software Reviews, Types of Review: Inspections, Deskchecks, Walkthroughs, Code Reviews			
Unit No 4	Software Configuration and Risk Management	Hours 9	4	AP
	Software Configuration Items and Tasks, Baselines, Plan for Change, Change Control, Change Requests Management, Version Control, Risk Management: Risks and Risk Types, Risk Breakdown Structure (RBS), Risk Management Process: Risk Identification, Risk Analysis, Risk Planning, Risk Monitoring, Cost Benefit Analysis, Software Project Management tools: CASE Tool, MS-Project.			
Unit No 5	Software Quality Assurance	Hours 9	5	AN
	Concept of Software Quality, Software Quality Attributes, Software Quality Metrics and Indicators, The SEI Capability Maturity Model (CMM), SQA Activities, Formal SQA Approaches: Proof of Correctness, Statistical Quality Assurance, Product versus process quality management, Introduction, types of contract, stages in contract, placement, typical terms of a contract, contract management, acceptance			

Textbooks

1	Bob Hughes, Mike Cotterell and Rajib Mall, "Software Project Management", 5 th Edition, Tata McGraw Hill, 2018, ISBN 13 978-0-07-710989-9.
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Reference Books / Journal Articles / Weblink

1	Roger Pressman, "A practitioner's Guide to Software Engineering", 8 th Edition, Tata McGraw Hill, 2014
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Software Project Management Tutorial

Purpose: To provide the basic skills and knowledge needed to effectively manage a group project. **Importance:** Project scheduling and understanding/handling risk is crucial to success in science and engineering.

Goals: Files: After this activity, you will be able to:

- Break a complex project into manageable sub-projects
- Assign responsibility/ownership of project components
- Set reasonable milestone's for the completion of major tasks
- Perform a risk analysis and integrate mitigation into your planning
- Develop and complete a project plan

SECTION 1: INTRODUCTION TO PROJECT MANAGEMENT
 SECTION 2: STAGES OF A PROJECT AND HOW TO COMPLETE THEM
 SECTION 3: PROJECT MANAGEMENT IN OUR PROJECT ACTIVITIES.
 SECTION 4: ACTIVITY - START YOUR PROJECT

BTECCE23H508:Network Security

Course Type	Honor	Semester	5
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Teaching Scheme		Credits		Examination Scheme	
Lecture	3 Hr./Week	Lecture	3	CIE Marks	60
Tutorial	-- Hr./Week	Tutorial	-	ESE Marks	40
Practical/Studio	0 Hr./Week	Practical/Studio	0	Total Marks	100
Total	3 Hr./Week	Total	3		

Course Description

This course helps to students to learn the algorithms/mechanisms for implementing security solutions. Identifies computer and network security threats, classify the threats and develop a security model to prevent, detect and recover from the attacks. Analyzes existing authentication and key agreement protocols, identify the weaknesses.

Course Outcomes

CO No.	Statement
1	Explain the foundational theory behind network security.
2	Learn various attacks that may occur inside network
3	Encrypt and decrypt messages using block ciphers, sign and verify messages using well known signature generation and verification algorithms.
4	Use the algorithms/techniques to achieve confidentiality and information integrity in information systems.
5	Use authentication and key agreement protocols in the real-world applications/systems.

Mapping of COs to POs and PSOs

CO No	POs												PSOs				BTL
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4	
CO1			2														UN
CO2				2													AP
CO3						2											AN
CO4								3									AP
CO5															3		AN

Affinity Level: 1- Slight, 2- Moderate, 3-Substantial, BTL: Bloom's Taxonomy Level

Course Content

Unit No 1	Security Fundamentals	Hours9	CO	BT L
Need for security, Security Goals, Cryptographic Attacks, Services and Mechanisms, Techniques for security goals implementations. Classical Encryption Techniques-: Symmetric Cipher Model, Substitution Techniques, Transposition Techniques, Steganography.			1	UN
Unit No 2	Network Security Threats, Vulnerabilities and Attacks	Hours 9	2	AP
Essential Terminologies , Network Security Concerns ,Why Network Security Concerns Arise Types of Network Security Threats ,Different Types of network security threats, How does a Network Security Breach Affect Business Continuity, Types of Network Security Vulnerabilities: Technological,Types of Network Security Vulnerabilities:Security Policy. Types of Network Security Attacks,Reconnaissance Attacks ,Reconnaissance Attacks:ICMP Scanning,Reconnaissance Attacks:DNS Footprinting, Reconnaissance attacks:Network InformationExtraction using Nmap, Reconnaissance Attacks: Port Scanning Reconnaissance Attacks: Social Engineering , Access Attacks : Password Attacks, Password Attack Techniques , Access Attacks : Network Sniffing , Access Attacks: Man in the Middle Attack ,Access Attacks : Replay Attack , Access Attacks : Privilege Escalation Access Attacks: DNS Poisoning, Access Attacks: DNS Cache Poisoning Access Attacks: ARP Poisoning, Access Attacks : DHCP Starvation Attacks Access Attacks : DHCP Spoofing Attack, Access Attacks : DHCP Spoofing Attack(CONT'D) Access Attacks : Switch Port Stealing, Access Attacks : MAC Spoofing/Duplicating Denial of Service Attack (DOS), Distributed Denial of Service Attack:(DDoS), Malware Attacks.				
Unit No 3	Symmetric Ciphers and Asymmetric Ciphers	Hours9	3	AN
Block Cipher Principles, The Data Encryption Standard (DES), A DES Example, The Strength of DES, Differential and Linear Cryptanalysis, Block Cipher Design Principles, The Origins AES, AES Structure, AES Round Functions, AES Key Expansion, An AES Example, AES Implementation. Block Cipher Operation: Multiple Encryption and Triple DES, Electronic Codebook Mode, Cipher Block Chaining Mode, Cipher Feedback Mode, Output Feedback Mode. Principles of Public-Key Cryptosystems, The RSA Algorithm, Diffie-Hellman Key Exchange, Elliptic Curve Arithmetic, Elliptic Curve Cryptography, Pseudorandom Number Generation Based on an Asymmetric Cipher.				
Unit No 4	Cryptographic Data Integrity Algorithms	Hours 9	4	AP
Applications of Cryptographic Hash Functions, Secure Hash Algorithm (SHA), Message Authentication Codes, Security of MACs, MACs Based on Hash Functions: HMAC, Authenticated Encryption: CCM and GCM, Digital Signatures				
Unit No 5	Mutual Trust and Internet Security	Hours9	5	AN

Key Management and Distribution: Distribution of Public Keys, X.509 Certificates, Public Key Infrastructure, Remote User Authentication Principles, Remote User Authentication Using Symmetric Encryption, Remote User Authentication Using Asymmetric Encryption Pretty Good Privacy (PGP), S/MIME, Domain Keys Identified Mail (DKIM), Internet Key Exchange, Cryptographic Suites. Blockchain: Blockchain Introduction, Hashing, Markel Tree, Proof Of Work Method, Consensus Introduction, Decentralized Blockchain Network – Introduction.		
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Textbooks

1	<i>William Stallings, “Cryptography and Network Security Principles and Practice”, 5th Edition, Pearson, 2011, ISBN 10:0-13-609704-9 [Recommended]</i>
2	<i>Behrouz Forouzan, “Cryptography and Network Security”, McGraw Hill.</i>

Reference Books / Journal Articles / Weblink

1	<i>Ninad Godbole, “Information Systems Security”, Wiley India Pvt. Ltd, ISBN -978-81-265-1692-6.</i>
2	<i>Bernard Menezes, “Network Security and Cryptography”, Cengage Learning, ISBN-978-81-315-1349-1</i>
3	<i>Mark Merkow, “Information Security-Principles and Practices”, Pearson Ed. 978-81-317-1288-7.</i>

SEM VI

BTECCE23601:: Artificial Intelligence

Course Type	PCC	Semester	6
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Teaching Scheme		Credits		Examination Scheme	
Lecture	3Hr./Week	Lecture	3	CIE Marks	60
Tutorial	--Hr./Week	Tutorial	-	ESE Marks	40
Practical/Studio	0Hr./Week	Practical/Studio	0	Total Marks	100
Total	3Hr./Week	Total	3		

Course Description

This course provides an in-depth understanding of Artificial Intelligence (AI), covering intelligent agents, search algorithms, logic representation, Natural Language Processing (NLP), Generative AI and AI ethics. It integrates hands-on experience with industry-standard tools such as TensorFlow, PyTorch, OpenAI frameworks, and Scikit-learn.

Course Outcomes

CONo.	Statement
1	Understand the fundamental concepts of AI and intelligent agents.
2	Apply and evaluate search algorithms and optimization techniques.
3	Implement AI models for real-world problem-solving using Machine Learning frameworks.
4	Analyze and apply Natural Language Processing techniques for AI applications.
5	Develop generative AI models and assess ethical considerations in AI applications.

Mapping of COs to POs and PSOs

CO No	POs												PSOs				BTL
	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PS O1	PSO 2	PSO 3	PSO 4	
CO1	2																UN
CO2		3															AP
CO3				3													AN
CO4						3											AP
CO5																3	AN

Affinity Level: 1- Slight, 2- Moderate, 3-Substantial, BTL: Bloom's Taxonomy Level

Course Content

Unit No 1	Introduction to AI & Intelligent Agents	Hours 9	CO	BTL
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Introduction: Introduction to Artificial Intelligence, Non-AI(Genetic Algorithm, Hidden Markov Model), AI Techniques(Deep Learning, ANN), Weak AI Vs Strong AI Intelligent Agents: Agents and Environments, concept of rationality, the nature of environments, Types and Structures of Agents. Problem Formulation: Vacuum World, 8 Queens, Route Finding, Robot Navigation Application of AI in Industry - AI in Healthcare, Finance, Self-Driving Cars, and Cloud Computing- AI for Business Intelligence & MLOps Tools: OpenAI, Google Colab, Python (NumPy, Pandas) Case Study: AI vs. Traditional Systems (HackerRank, UpGrad)			1	UN
Unit No 2	Search Strategies & Optimization	Hours 9	2	AN
Introduction - Search Techniques Uninformed Search: DFS, BFS, Depth-Limited, Iterative Deepening, Bidirectional Search Informed Search: A*, AO*, Heuristic-based Search Optimization Techniques: Genetic Algorithms, Hill Climbing, Constraint Satisfaction Game Playing: Minimax, Alpha-Beta Pruning Tools: Scikit-learn, NetworkX for Graph-based Search Case Study: Search Space Optimization in AI-based Robotics (Coursera)				
Unit No 3	Knowledge Representation & Expert Systems	Hours 9	3	AP
Introduction to Knowledge Representation Logical Agents: Knowledge based agents, Wumpus world. Propositional Logic: Representation, Inference, Reasoning Patterns, Resolution, Forward and Backward Chaining. First order Logic: Representation, Inference, Reasoning Patterns, Resolution, Forward and Backward Chaining. Expert System: Design and Implementation - Tools: Prolog, SWI-Prolog, LISP - Case Study: AI-powered Legal Reasoning (NPTEL)				
Unit No 4	Natural Language Processing (NLP)	Hours 9	4	AN
Introduction to Natural Language Processing: Steps of NLP Text Processing: Tokenization, Lemmatization, Stop word Removal Syntactic Analysis: POS Tagging, Parsing, Tree Representation Semantic Processing: Named Entity Recognition, Word Embeddings (Word2Vec, GloVe), Sentiment Analysis, Chatbots, Language Modeling Tools: NLTK, SpaCy, Hugging Face Transformers, BERT Case Study: AI-based Chatbots in Customer Service (UpGrad, Coursera)				
Unit No 5	Generative AI & Ethical Considerations	Hours 9	5	AP
Text & Image Generation: GANs, Diffusion Models, DALL-E, ChatGPT Conversational AI: Chatbot Development with GPT-4, Gemini Explainable AI: Bias Detection & Model Interpretability AI Ethics: Privacy, Fairness, AI Bias, Societal Impact Tools: OpenAI GPT, Google Gemini, Explainable AI (XAI) Frameworks Case Study: Ethical AI in Healthcare (Simplilearn, NPTEL)				

Textbooks

1	Elaine Rich and Kevin Knight, "Artificial Intelligence" Tata McGraw Hill
2	Stuart Russell & Peter Norvig, "Artificial Intelligence: A Modern Approach", Pearson Education, 2nd Edition.
3	Deepak Khemani, "A First Course in Artificial Intelligence", McGraw Hill Education (India) Pvt. Ltd

Reference Books / Journal Articles / Weblink

1	Nilsson: "Principles of Artificial Intelligence", Morgan Kaufmann.
2	Eugene, Charniak, Drew McDermott: "Introduction to Artificial Intelligence."
3	Patterson: "Introduction to AI and Expert Systems", PHI
4	NumaDhamani, Maggie Engler: "Introduction to Generative AI: An ethical, societal and legal overview, Manning Publications.
5	https://www.cmu.edu/intelligentbusiness/expertise

BTECCE23602 :: Artificial Intelligence Lab

Teaching Scheme		Credits		Examination Scheme	
Lecture	0Hr./Week	Lecture	0	CIE Marks	15
Tutorial	--Hr./Week	Tutorial	-	ESE Marks	10
Practical/Studio	2Hr./Week	Practical/Studio	1	Total Marks	25
Total	2Hr./Week	Total	1		

List of Experiments

1	Implement Non-AI Techniques for the following problems: a. Tic Tac Toe b. N Queens c. Magic square
2	Implement the Water Jug problem using the following uninformed search strategies: a. Depth First Search b. Breadth First Search
3	Implement the Hill Climbing technique to solve the 8 puzzle problem.
4	Implement the Best First Search to solve the following problems: a. 8 puzzle b. Robot Navigation problem c. Cities Distance (shortest path) problem
5	Implement the A* algorithm to solve the following problems given by HackerRank: a. N puzzle b. Robot Navigation problem c. Cities Distance (shortest path) problem
6	Implement Constraint Satisfaction Algorithm for the following problems: a. Cryptarithmic b. Crossword puzzle c. Map coloring problem
	Implement the Minimax algorithm to solve the Tic Tac Toe problem
8	Implement NLP problems statements by Hacker Rank for following: a. POS Tagging b. Similarity Score c. Spell Checker
9	Implement a comparative study and analyze the differences between ChatGPT and Google Gemini models.

BTECCE23603:: Statistical Data Analysis and Visualization

Course Type	PCC	Semester	6
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Teaching Scheme		Credits		Examination Scheme	
Lecture	3 Hr./Week	Lecture	3	CIE Marks	60
Tutorial	0 Hr./Week	Tutorial	0	ESE Marks	40
Practical/Studio	0Hr./Week	Practical/Studio	0	Total Marks	100
Total	3 Hr./Week	Total	3		

Course Description

Statistical Data Analysis and Visualization is a fundamental course designed to provide students with a comprehensive understanding of statistical techniques and their applications in analyzing and interpreting data. The course emphasizes the practical aspects of statistical analysis and equips students with the necessary skills to make informed decisions based on data.

Course Outcomes:

CO	Statement
1	Gain an understanding of statistical methods relevant to upper division interdisciplinary courses.
2	Sharpen students' statistical intuition and abstract reasoning as well as their reasoning from numerical data through community-based and other research.
3	Learn and apply the procedure of hypothesis testing.
4	Understand the relationships within data using regression and correlation.
5	Learn advance techniques for statistical analysis.

Mapping of COs to POs and PSOs

CO No	P O 1	PO 2	POs										PSOs				BT L
			PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	P O 1 0	P O 1 1	P O 1 2	PSO 1	PSO 2	PSO 3	PSO 4	
CO1	3																UN
CO2		2															AP

CO3				3													AP
CO4					2												AP
CO5															3		AN

Affinity Level: 1- Slight, 2- Moderate, 3-Substantial, BTL: Bloom's Taxonomy Level

Course Contents

UNIT-1	Introduction to Statistics - Organizing and Graphing Data	Hours 9	CO	BTL
	What is statistics? - Types of statistics - Population versus sample - Basic terms - types of variables - cross section verses time series data - different types of data - sources of data - designing studies and collecting good quality data Raw data- organizing and graphing qualitative data - organizing and graphing quantitative data - Histograms - Cumulative Frequency Distributions - Stem-and-Leaf Display		1	UN
UNIT-2	Numerical Descriptive Measures	Hours 9	2	AP
	Measures of central tendency for ungrouped data -Measures of dispersion for ungrouped data - Mean, variance, and standard deviation for grouped data - Use of standard deviation - Measures of position - Box-and-Whisker Plot			
UNIT-3	Discrete Random Variables and Their Probability Distributions	Hours 9	3	AP
	Random variables -probability distribution of a discrete random variable -mean of a discrete random variable - standard deviation of a discrete random variable - The binomial probability distribution - The Poisson probability distribution			
UNIT-4	Continuous Random Variables & Normal Distribution-Sampling Distributions	Hours 9	4	AP
	Continuous probability distribution -The normal distribution - The standard normal distribution - Standardizing a normal distribution - Applications of the normal distribution - determining the z and x values when an area under the normal distribution curve is known -The normal approximation to the binomial distribution. Population and sampling distributions- mean and standard deviation of $\bar{x}$ - applications of the sampling distribution of $\bar{x}$.			
UNIT-5	Estimations and Hypothesis Testing	Hours 9	5	AN
	Point and interval estimates - estimation of a population mean: (σ known)-estimation of a population mean: (σ not known) Hypothesis tests: Introduction -Hypothesis tests about population mean (σ known) - Hypothesis tests about population mean (σ not known) Chi-Square Tests - The Chi-square distribution - Goodness-of-fit test - Contingency tables - Test of independence or homogeneity - Inferences about the population variance Analysis of Variance - The F Distribution - One-Way Analysis.			

Text Books

1	Mann Prem S., Lacke Christopher Jay. (2010). Introductory Statistics(Jenn Albanese, Ellen Keohane),7, John Wiley & Sons, Inc
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2	<i>Christopher Chatfield. (1994). Statistics for Technology: A Course In Applied Statistics, 3, Taylor & Francis Group.</i>
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Reference Book

1	<i>Bruce Peter, Bruce Andrew. (2017). Practical Statistics for Data Scientists(Shannon Cutt), I, O'Reilly Media, Inc.</i>
2	<i>Easterling Robert G. (2015). Fundamentals of Statistical Experimental Design and Analysis(Prachi Sinha Sahay, Jo Taylor, Kathryn Sharples), John Wiley & Sons, Ltd.</i>

BTECCE23604 :: Statistical Data Analysis and Visualization Lab

Teaching Scheme		Credits		Examination Scheme	
Lecture	0Hr./Week	Lecture	0	CIE Marks	15
Tutorial	--Hr./Week	Tutorial	-	ESE Marks	10
Practical/Studio	2Hr./Week	Practical/Studio	1	Total Marks	25
Total	2Hr./Week	Total	1		

List of Experiments

1	Introduction to R Language and basic R commands- basic calculations - help - installing packages -datatypes - data structures - vectors - subsetting - logical operators -matrices - calculations with vectors and matrices - lists - data frames
2	Programming in R - control flow - if/else - for loop - functions - Reading in data from different sources
3	Summarizing Data - Use R commands to find following summery measures of the given 3 datasets- mean , median , mode, standard deviation, range, variance, coefficient of variation, minimum, maximum, quartiles , interquartile range (IQR). Compare the data-sets based these descriptive measures.-- Data cleaning -missing values
4	Graphing data - Plot following graphs - frequency distribution, bar-plots, pie chart, histogram, polygon and box-and-whisker plot, scatter-plots, stem-and-leaf plot, (Examples of continuous and discrete data)
5	Probability Distributions - Computing probabilities of an event by applying various probability distribution function viz. binomial distribution , Poisson distribution , normal distribution , Chi-squared distribution , student t distribution , F distribution, Confidence interval computation
6	Hypothesis testing -1 - One sample t-Test - Two sample t-Test , independent samples t-test - non-parametric tests - One-way ANOVA - Two-way ANOVA
7	Hypothesis testing -2 - Comparing group means using the independent samples t test , F-test , Chi-square test ,
8	Confidence Interval Estimation - Proportion test , z-test , t-test , confidence

BTECCE23605A:: Machine Learning

Course Type	PEC	Semester	7
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Teaching Scheme		Credits		Examination Scheme	
Lecture	3Hr./Week	Lecture	3	CIE Marks	60
Tutorial	--Hr./Week	Tutorial	-	ESE Marks	40
Practical/Studio	0Hr./Week	Practical/Studio	0	Total Marks	100
Total	3Hr./Week	Total	3		

Course Description

This course helps the student to identify the role of Supervised, Unsupervised and Reinforcement Algorithms. Illustrates concept learning and formulate research-based problems based on Machine Learning techniques. Analyzes different algorithms used in Machine Learning as applied to generic datasets.

Course Outcomes

CO No.	Statement
1	Demonstrate knowledge of Supervised, Unsupervised and Reinforcement algorithms through implementation for sustainable solutions of applications.
2	Evaluate different classification algorithms through implementation for sustainable solutions of applications.
3	Apply different clustering algorithms used in Machine Learning to generic datasets and specific multidisciplinary domains.
4	Analyze research-based problems using Neural Networks and genetic algorithms.
5	Evaluation of different convolution neural network algorithms on well formulated problems along with stating valid conclusions that the evaluation supports.

Mapping of COs to POs and PSOs

CO No	POs												PSOs				BT L
	P O 1	P O 2	P O 3	PO 4	PO 5	P O 6	PO 7	P O 8	P O 9	PO 10	PO 11	PO 12	PS O1	PS O2	PS O3	PS O4	
CO1		3															UN
CO2				3													AP
CO3					2												AN
CO4							2										AP
CO5																3	AN

Affinity Level: 1- Slight, 2- Moderate, 3-Substantial, BTL: Bloom's Taxonomy Level

Course Content

Unit No 1	Introduction	Hours 9	CO	BTL
Introduction: What is Machine Learning? Types of learning: Supervised, Unsupervised, Reinforcement. Simple regression: Types, Making predictions, Cost function, Gradient descent, Training, Model evaluation. Multivariable Regression: Growing complexity, Normalization, Making predictions, Initialize weights, Cost function, Gradient descent, Simplifying with matrices, Bias term, Model evaluation Learning System: Well posed learning problems, Designing a Learning system, Applications of machine learning, Perspective and Issues in Machine Learning			1	UN
Unit No 2	Classification Methods	Hours 9	2	AP
Support Vector Machine Classification Algorithm, Hyper plane, optimal separating Hyper planes, kernel functions, kernel selection, Applications Introduction to Ensemble and its techniques, Bagging and Bootstrap Ensemble methods Introduction to Random Forest, growing of Random Forest, Random feature selection				
Unit No 3	Clustering Methods	Hours 9	3	AP
Overview of Clustering and Unsupervised Learning Introduction to Clustering methods: Partitioning methods K-Means algorithm, assessing quality and choose number of clusters, KNN (1 NN, K NN) techniques, K-Medians, Density based method: Density-Based Spatial Clustering Hierarchical clustering methods: Agglomerative Hierarchical clustering technique, Roles of Dendrograms and Choosing number clusters in Hierarchical clustering, Divisive clustering techniques, Principal Component Analysis (PCA)				
Unit No 4	Neural network and Genetic Algorithms	Hours 9	4	AP
Neural Network Representation – Problems – Perceptron – Multilayer Networks and Back Propagation Algorithms – Advanced Topics – Genetic Algorithms – Hypothesis Space Search – Genetic Programming – Models of Evaluation and Learning.				
Unit No 5	Convolutional Neural Network	Hours 9	5	AN
Convolutional Neural Network Recursive Neural Network, Recurrent Neural Network, Long-short Term Memory, Gradient descent optimization				

Textbooks

1	<i>T. Mitchell, "Machine Learning", McGraw-Hill, 1997.</i>
2	<i>Douglas Montgomery, Elizabeth A. Peck, and G. Geoffrey Vining, "Introduction to Linear Regression Analysis", 5th edition, Wiley publication</i>

3	Charu C. Aggarwal, Chandan K. Reddy, "Data Clustering Algorithms and Applications".
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Reference Books / Journal Articles / Weblink

1	Ethem Alpaydin, "Introduction to Machine Learning", MIT press, 2004.
2	Jacek M. Zurada, "Introduction to Artificial neural System", JAICO publishing house, 2002.

BTECCE23606A :: Machine Learning Lab

Teaching Scheme		Credits		Examination Scheme	
Lecture	0Hr./Week	Lecture	0	CIE Marks	15
Tutorial	--Hr./Week	Tutorial	-	ESE Marks	10
Practical/Studio	2Hr./Week	Practical/Studio	1	Total Marks	25
Total	2Hr./Week	Total	1		

List of Experiments

1	Create a Regression model of your choice using a suitable dataset.
2	Create a Multi variable Regression model of your choice using a suitable dataset.
3	Write a program to implement Support Vector Machine Algorithm to solve classification problems.
4	Write a program to implement Boosting and Bagging methods. Use UCI Machine Learning repository dataset.
5	Write a program to implement K-nearest neighbor algorithm.
6	Write a program to implement K-means clustering algorithm.
7	Write a program to implement the single perceptron model.
8	Write a program implement backpropagation algorithm.
9	Implement CNN for image classification.

BTECCE23605B:: Internet of Things

Course Type	PEC	Semester	6
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Teaching Scheme		Credits		Examination Scheme	
Lecture	3 Hr./Week	Lecture	3	CIE Marks	60
Tutorial	-- Hr./Week	Tutorial	-	ESE Marks	40
Practical/Studio	0 Hr./Week	Practical/Studio	0	Total Marks	100
Total	3 Hr./Week	Total	3		

Course Description

This course will help students to learn the concepts of IoT and different applications in IoT.

Course Outcomes

CO No.	Statement
1	Explain in a concise manner how the general Internet as well as IoT works.
2	Understand constraints and opportunities of wireless and mobile networks for Internet of Things.
3	Choose between available technologies and devices for stated IoT challenge.
4	Evaluation of IoT Security models
5	Analyze case studies of IoT systems

Mapping of COs to POs and PSOs

CO No	POs												PSOs				BTL
	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3	PSO 4	
CO1			2														UN
CO2					3												AP
CO3						2											AP
CO4							3										AP
CO5															3		UN

Affinity Level: 1- Slight, 2- Moderate, 3-Substantial, BTL: Bloom's Taxonomy Level

Unit No 1	Basic Concepts Revisited	Hours 9	CO	BTL
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Introduction to sensing & actuating, Basic networking, Wireless networks, Wireless sensor networks (WSN), Communication protocols, and other enabling technologies, IoT standards, Data storage & management issues and approaches, Cloud computing, Key challenges, and future directions of IoT, and security & privacy issues.			1	UN
Unit No 2	Embedded Systems	Hours 9	2	AP
Hardware and software of IoT, Microcontrollers, Understanding and programming Arduino, Raspberry Pi, NodeMCU, Lora, etc. Integrating microcontrollers with sensors and actuators, Building the IoT applications with any microcontroller.				
Unit No 3	IoT Architectures and Protocols	Hours 9	3	AP
State-of-the-art, IoT architecture reference models, Different views of IoT architectures and frameworks design, Protocols: Application protocols, Service discovery protocols, Infrastructure protocols, and other protocols. Understanding various types of protocols like HTTP, MQTT, CoAP, AMQP, 6LoWPAN, etc. Cross-layer implementations, and Data dissemination.				
Unit No 4	Support Technologies for IoT	Hours 9	4	AP
Big Data, Data Analytics, Artificial Intelligence, Mobile, Cloud, Software defined networks, 5G, and Fog/Edge computing. IoT integration with recent technologies. State-of-the-art. Design goals, challenges, and components.				
Unit No 5	IoT Security & Privacy	Hours 9	5	AN
IoT Security and Privacy issues and challenges, Risks involved with IoT infrastructures, Trust in IoT platforms and other integrating technologies, Data aggregation, storage, retrieval, and other management issues including fault tolerance, interoperability, security, and privacy, Cyber-physical systems and their security and privacy, Mitigation approaches.				

Textbooks

1	Arshdeep Bahga, Vijay Madisett, "Internet of Things – A hands-on approach", Universities Press, ISBN: 0: 0996025510, 13: 978-0996025515.
2	Honbo Zhou, "The Internet of Things in the Cloud: A Middleware Perspective", CRC Press, 2012. ISBN: 9781439892992.
3	Dieter Uckelmann, Mark Harrison, Florian Michahelles, "Architecting the Internet of Things", Springer, 2011. ISBN: 978-3-642-19156-5.
4	Lyla B. Das, "Embedded Systems: An Integrated Approach", Pearson, ISBN: 9332511675, 9789332511675.

Reference Books / Journal Articles / Weblink

1	David Easley, Jon Kleinberg, "Networks, Crowds, and Markets: Reasoning About a Highly Connected World", Cambridge University Press, 2010, ISBN: 10: 0521195330.
2	Olivier Hersent, Omar Elloumi, David Boswarthick, "The Internet of Things: Applications to the Smart Grid and Building Automation", Wiley, 2012, 9781119958345.
3	Olivier Hersent, David Boswarthick, Omar Elloumi, "The Internet of Things – Key applications and Protocols", Wiley, 2012, ISBN: 978-1-119-99435-0.

4	<i>Barrie Sosinsky, "Cloud Computing Bible", Wiley-India, 2010.ISBN: 978-0-470-90356-8.</i>
5	<i>Adrian McEwen, Hakim Cassimally, "Designing the Internet of Things", Wiley, 2014, ISBN: 978-1-118-43063-7.</i>
6	<i>Christopher Hallinan, "Embedded Linux Primer", Prentice Hall, ISBN: 13: 978-0-13-167984-9.</i>

BTECCE23606B :: Internet of Things Lab

Teaching Scheme		Credits		Examination Scheme	
Lecture	0Hr./Week	Lecture	0	CIE Marks	15
Tutorial	--Hr./Week	Tutorial	-	ESE Marks	10
Practical/Studio	2Hr./Week	Practical/Studio	1	Total Marks	25
Total	2Hr./Week	Total	1		

Mini Project

Students have to implement mini projects which should include following concepts:

1	Node Structure: Sensing, Processing, Communication, Powering
2	Networking: Topologies, Layer/Stack architecture
3	Communication Technologies: Introduction to ZigBee, BLE, WiFi, LTE, IEEE 802.11ah, Discuss data rate, range, power, computations/bandwidth, QoS
4	Smartness - Signal Processing/Analytics: Impact on Power/Energy savings, dynamic networks, simple case studies
5	IoT Fabricator: Introduction to Embedded electronics, fabricating electronics, Communication Network requirements, Data processing challenges – recreation, IP/security, Challenges
6	Hands-on in IoT: Projects based on some Hardware (Raspberry pi, Arduino, Intel, IITH Mote, Smartphone), Software (Contiki, TinyOS, Android), IoT Fabricator etc. can be used.

BTECCCE23607A: Ethical Hacking and Penetration Testing

Course Type	PEC	Semester	6
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Teaching Scheme		Credits		Examination Scheme	
Lecture	3Hr./Week	Lecture	3	CIE Marks	60
Tutorial	--Hr./Week	Tutorial	-	ESE Marks	40
Practical/Studio	0Hr./Week	Practical/Studio	0	Total Marks	100
Total	3Hr./Week	Total	3		

Course Description

This course helps the student to gain the knowledge of Ethical Hacking from a vendor-neutral perspective. Identifies vulnerability assessment and penetration testing methodology and techniques. Understand concepts of VPN, Network Security Policy Design and Implementation and wireless network defense mechanisms.

Course Outcomes

CO No.	Statement
1	Able to identify vulnerabilities and weakness.
2	Able to perform penetration testing.
3	Able to perform wireless network attacks and vulnerabilities identification.
4	Able to perform firewall, IDS evasion.
5	Able to identify vulnerabilities and attacks in web servers.

Mapping of COs to POs and PSOs

CO No	POs											PSOs				BTL
	P O 2	PO 3	PO 4	P O 5	PO 6	P O 7	PO 8	P O 9	P O 10	P O 11	P O 12	PS O1	PS O2	PS O3	PS O4	
CO1		2														UN
CO2			2													AP
CO3					3											AP
CO4							3									AP
CO5														3		AN

Affinity Level: 1- Slight, 2- Moderate, 3-Substantial, BTL: Bloom's Taxonomy Level

Course Content

Unit No 1	Introduction to Ethical Hacking	Hours10	CO	BTL
	Information Security Threats and Attack Vectors, Hacking Concepts: Types and Phases (Reconnaissance, Scanning, Gaining Access, Maintaining Access, Covering Tracks), Ethical Hacking: Principles, Scope, and Responsibilities, Vulnerability Assessment vs. Penetration Testing, Information Security Laws and Standards (ISO 27001, GDPR, HIPAA, etc.), Footprinting Techniques (Passive and Active), Network Scanning Methodologies, Enumeration Techniques: NetBIOS, SNMP, and SMTP Enumeration, Enumeration Countermeasures Industry Tools: Nmap – Network scanning and discovery, Recon-ng – Footprinting and reconnaissance, Maltego – Link analysis and intelligence gathering		1	UN
Unit No 2	Vulnerability Assessment and Penetration Testing (VAPT)	Hours10	2	AP
	Understanding Vulnerability Assessment and its Life-Cycle,,Identifying CERT Alerts and Resource ThreatsVulnerability Scanning vs. Penetration Testing, Automated Security Scanners, Web Application Exploitation:Understanding the Attack Surface, Identifying and Reporting Vulnerabilities,Logical Flaws in Web AppsHacking Demonstration and Documentation, Industry Tools: OpenVAS / Greenbone – Vulnerability scanningNessus – Professional vulnerability scanner, Burp Suite – Web application penetration testing OWASP ZAP – Open-source web vulnerability scanner, Metasploit Framework – Penetration testing			
Unit No 3	Malware Threats and Social Engineering	Hours8	3	AP
	Malware Types: Viruses, Worms, Trojans, Ransomware, Malware Reverse Engineering (Basics) ,Social Engineering Techniques, Impersonation, Pretexting, Phishing, and Baiting, Identity Theft Techniques,Session Hijacking: Techniques (e.g., MITM, Cookie Hijacking), Types and Prevention Industry Tools:,Wireshark – Network packet analyzer Social Engineer Toolkit (SET) – Phishing and impersonation,Volatility – Memory forensics PESTudio – Static malware analysis IDA Pro / Ghidra – Reverse engineering			
Unit No 4	Evading IDS, Firewall and Honeypots	Hours8	4	AP
	Intrusion Detection Systems (IDS), Firewalls, and Honeypots – Architecture and Operation, Evasion Techniques for IDS/IPS, Evading Stateful Firewalls, Honeypot Detection Techniques, Countermeasures and Defense Mechanisms, Industry Tools: Hping3 – Packet crafting and firewall evasion, Nmap NSE Scripts – IDS evasion Snort / Suricata – Open-source IDS/IPS KFSensor / Honeyd / Cowrie – Honeypots Scapy – Packet manipulation			
Unit No 5	Hacking Web Servers and Wireless Network Penetration Testing	Hours9	5	AN
	Introduction to Bug Bounty, Definition, scope, and real-world relevance, Role of ethical hackers and responsible disclosure, Overview of Vulnerability Disclosure Programs (VDPs & RDPs), Beginner roadmap: tools, labs, and learning resources, Understanding OWASP Top 10 vulnerabilities, Legal and ethical considerations in bug hunting, Bug Bounty Platforms, Google Dorking, Shodan, WHOIS, DNS enumeration, Finding leaked data on GitHub and Pastebin, System Hacking Fundamentals, Gaining initial access and password cracking basics (John, Hashcat), Privilege escalation tactics on Windows & Linux, Post-			

exploitation: credential dumping, persistence methods		
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Textbooks

1	<i>IT Network security (IBM ICE Publication)</i>
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Reference Books / Journal Articles / Weblink

1	<i>Certified Ethical Hacking, EC-Council Study Guide.</i>
2	<i>Ninad Godbole, "Information Systems Security", Wiley India Pvt. Ltd, ISBN -978-81-265-1692-6.</i>
3	<i>Mark Merkow, "Information Security-Principles and Practices", Pearson Ed. 978-81-317-1288-7.</i>
4	<i>William Stallings, "CRYPTOGRAPHY AND NETWORK SECURITY PRINCIPLES AND PRACTICE", FIFTH EDITION, PEARSON, 2011, ISBN 10:0-13-609704-9</i>
5	<i>BehrouzForouzan, "Cryptography and Network Security", McGraw</i>
6	https://github.com/patilkr?tab=repositories

BTECCCE23608A: Ethical Hacking and Penetration Testing Lab

Teaching Scheme		Credits		Examination Scheme	
Lecture	0Hr./Week	Lecture	0	CIE Marks	15
Tutorial	--Hr./Week	Tutorial	-	ESE Marks	10
Practical/Studio	2Hr./Week	Practical/Studio	1	Total Marks	25
Total	2Hr./Week	Total	1		

List of Experiments

1	Footprinting a target network
2	Network Scanning
3	Enumeration Tools
4	Social Engineering using Kali Linux
5	Sniffing the network using various tools
6	Password Cracking
7	Hacking web Server/Web server auditing
8	Introduction to BurpSuite, Covering Tracks: Clearing Logs
9	Hacking Wi-Fi Protected Access Network
10	Evading IDS, Firewall and honeypots
11	Vulnerability Scanning using Nessus Vulnerability Scanning Tool

BTECCE23607B:: Real Time Operating System

Course Type	PEC	Semester	6
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Teaching Scheme		Credits		Examination Scheme	
Lecture	3Hr./Week	Lecture	3	CIE Marks	60
Tutorial	--Hr./Week	Tutorial	-	ESE Marks	40
Practical/Studio	0Hr./Week	Practical/Studio	0	Total Marks	100
Total	3Hr./Week	Total	3		

Course Description

This course explores the fundamentals of Operating Systems (OS) and Real-Time Operating Systems (RTOS), covering OS architecture, process management, concurrency, scheduling, and memory management. It introduces real-time concepts, scheduling algorithms, and RTOS kernel design, emphasizing deterministic behavior. Students will analyze multithreading, synchronization mechanisms, and inter-process communication. Through theoretical insights and hands-on activities, students will develop a strong foundation in OS and RTOS concepts, preparing them for real-time and embedded system development in critical domains. The course includes real-world RTOS case studies.

Course Outcomes

CO No.	Statement
1	Understand the fundamentals of operating systems, including mobile OS like Android.
2	Analyze process management techniques, threading models, concurrency issues, and synchronization mechanisms in operating systems.
3	Demonstrate an understanding of RTOS concepts, real-time design issues, hardware considerations, and kernel architecture.
4	Evaluate real-time scheduling algorithms, memory management strategies, and process execution in RTOS environments.
5	Analyze real-world RTOS case studies in real world domains.

Mapping of COs to POs and PSOs

CO No	POs												PSOs				BT L
	P O 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PS O 1	PSO 2	PSO 3	PSO 4	
CO1	3																UN
CO2		2															AP

CO3					3												AN
CO4							2										AP
CO5																3	AN

Affinity Level: 1- Slight, 2- Moderate, 3-Substantial, BTL: Bloom's Taxonomy Level

Course Content

Unit No 1	Introduction	Hours 9	CO	BT L
	Basics of operating systems: Generations, Types, Structure, Services, System Calls, System Boot, System Programs, Protection and Security. Introduction to Process: Process Concepts, Process States, Process Control Block.		1	UN
Unit No 2	Process Management	Hours 9	2	AP
	Process management: Scheduling-Criteria, Scheduling Algorithms and their Evaluation, Threads, Threading Issues. Process synchronization: Introduction, Critical-Section Problem, Synchronization Hardware, Semaphores, Classic Problems of Synchronization, and Monitors, Deadlock Handling. Concurrency: Issues with concurrency, Principles of Concurrency			
Unit No 3	Basics of Real-Time OS	Hours 9	3	AN
	Terminology: RTOS concepts and definitions, real-time design issues, examples, Hardware Considerations: logic states, CPU, memory, I/O, Architectures, RTOS building blocks, Real-Time Kernel			
Unit No 4	Task Scheduling and Memory Management	Hours 9	4	AP
	Real-Time Scheduling Algorithms, Rate Monotonic Scheduling (RMS), Earliest Deadline First (EDF), Least Laxity First (LLF), Time Management in RTOS, Timer services and clock ticks, Periodic vs. Aperiodic tasks, Pipes Memory Management, Process stack management, run-time buffer size, swapping, overlays, block/page management, replacement algorithms, real-time garbage collection.			
Unit No 5	Case Studies in RTOS	Hours 9	5	AN
	RTOS in Automotive Systems: Introduction, Challenges and Future Trends, RTOS in Aerospace & Defense, RTOS in Medical Devices: Introduction, RTOS in PaceMakers and MRI devices, Challenges and Future Trends.			

Textbooks

1	"Operating Systems", Stalling William, Pearson Education, ISBN: 0-13-031999-6, 5th Edition.
2	"Operating System Concepts", Silberschatz A., Galvin P., Gagne G., John Wiley and Sons, ISBN: 9971-51-388-9, 6th Edition.
3	J. J Labrosse, "MicroC/OS-II: The Real-Time Kernel", Newnes, 2002.
4	Jane W. S. Liu, "Real-time systems", Prentice Hall, 2000.

Reference Books

1	"Modern Operating Systems", Tanenbaum Andrew S., Prentice Hall India, ISBN: 81-7808-447-3, 2nd Edition.
2	W. Richard Stevens, "Advanced Programming in the UNIX®

	Environment”, 2nd Edition, Pearson Education India, 2011.
3	Philips A. Laplante, “Real-Time System Design and Analysis”, 3rd Edition, John Wley& Sons, 2004
4	Doug Abbott, “Linux for Embedded and Real-Time Applications”, Newnes, 2nd Edition, 2011.

BTECCE23608B:: Real Time Operating System Lab

Teaching Scheme		Credits		Examination Scheme	
Lecture	0Hr./Week	Lecture	0	CIE Marks	15
Tutorial	--Hr./Week	Tutorial	-	ESE Marks	10
Practical/Studio	2Hr./Week	Practical/Studio	1	Total Marks	25
Total	2Hr./Week	Total	1		

List of Experiments

1	Execution of basic and advanced Unix commands
2	Create multiple tasks and observe FreeRTOS task switching. Demonstrate preemptive and cooperative scheduling.
3	Implement Binary and Counting Semaphores to synchronize two tasks where one task generates numbers, another task prints them.
4	Implement producer-consumer communication using FreeRTOS queues where one task writes messages, another reads and displays them.
5	Implement Rate Monotonic Scheduling (RMS) algorithm.
6	Implement Earliest Deadline First (EDF) algorithm.
7	Implement Least Laxity First (LLF) algorithm.
8	Simulate a traffic signal system using RTOS tasks and scheduling.
9	Implement an elevator control system where different tasks handle floor requests.
10	Create a simple text-based chat system where multiple tasks handle incoming messages.

BTECCE23609:: Advanced Web and Server Technology

Course Type	VSEC	Semester	6
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Teaching Scheme		Credits		Examination Scheme	
Lecture	1Hr./Week	Lecture	1	CIE Marks	50
Tutorial	--Hr./Week	Tutorial	-	ESE Marks	--
Practical/Studio	2Hr./Week	Practical/Studio	2	Total Marks	50
Total	3Hr./Week	Total	3		

Course Description

The course will enable students to understand the core concepts and techniques involved in web development and Spring Boot application development. Students will learn the fundamentals of client-side scripting using JavaScript and ReactJS, focusing on event handling, asynchronous programming, and web performance optimization. On the server side, students will develop skills in building RESTful APIs, integrating with databases using Spring Data JPA, and implementing error handling and validation in Spring Boot applications. The course will provide hands-on practice with modern web technologies, ensuring students can design, develop, and optimize full-stack applications for real-world use cases.

Course Outcomes

CO No.	Statement
1	Develop dynamic and interactive web applications using JavaScript, ReactJS, and modern ES6 features while optimizing for performance and user experience.
2	Understand the core concepts and setup of a Spring Boot application, including annotations and configuration files.
3	Build RESTful APIs using Spring Boot to handle HTTP requests and manage data communication effectively.
4	Develop data-driven applications using Spring Data JPA for efficient CRUD operations and custom database queries.
5	Apply exception handling and input validation techniques to ensure robust and user-friendly Spring Boot applications.

Mapping of COs to POs and PSOs

CO No	POs												PSOs				BT L
	P O 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PS O1	PSO 2	PSO 3	PS O4	
CO1	2																UN

CO2					3												AP
CO3									3								AN
CO4										2							AP
CO5															3		AN

Affinity Level: 1- Slight, 2- Moderate, 3-Substantial, BTL: Bloom's Taxonomy Level

Course Content

Unit No 1	Web development fundamentals and Client-Side Scripting	Hours 3	CO	BTL
JavaScript: Functions, Arrays, Objects, and Events;Asynchronous JavaScript: Callbacks, Promises, Introduction to ES6 Features: Arrow Functions, Template Literals;Basics of Fetch API for API Communication. ReactJS Fundamentals: Introduction to Components, JSX: State and Props Management Event Handling and Forms;Building a Simple SPA (Single Page Application) using React Basic Web Performance Optimization:Optimizing JavaScript for Faster Loading Image Optimization and Lazy Loading			1	UN
Unit No 2	Introduction to SpringBoot	Hours 3	2	AP
Overview of Spring Framework & Spring Boot Setting Up a Spring Boot Project (Spring Initializr) Understanding application.properties & application.yml Spring Boot Annotations (@SpringBootApplication, @Component, @Service, @Repository)				
Unit No 3	REST API Development	Hours 3	3	AN
Creating RESTful APIs with @RestController Handling GET, POST, PUT, DELETE Requests Path Variables & Request Parameters Request Body & Response Entity				
Unit No 4	Working with Databases	Hours 3	4	AP
Introduction to Spring Data JPA Configuring H2, MySQL, or PostgreSQL Defining Entities and Repositories CRUD Operations with JPA Using @Query and JPQL for Custom Queries				
Unit No 5	Exception Handling & Validation	Hours 3	5	AN
Global Exception Handling (@ControllerAdvice) Using @Valid for Input Validation Custom Error Messages				

Textbooks

1	"Learning React: Modern Patterns for Developing React Apps" by Alex Banks and Eve Porcello, O'Reilly Media, ISBN: 978-1492051725
2	"Spring Boot in Action" by Craig Walls, Manning Publications, ISBN: 978-1617292545
3	"RESTful Web APIs" by Leonard Richardson and Mike Amundsen, O'Reilly Media, ISBN: 978-1449358068

Reference Books / Journal Articles / Weblink

1	"Eloquent JavaScript" by Marijn Haverbeke offers an in-depth exploration of JavaScript fundamentals, including functions, objects, and asynchronous programming.
2	"Spring Boot: Up and Running" by Mark Heckler, O'Reilly Media, ISBN: 978-1492076988
3	"Pro Spring Boot 2: An Authoritative Guide to Building Microservices, Web and Enterprise Applications, and Best Practices" by Felipe Gutierrez, Apress, ISBN: 978-1484236767

4	https://www.researchgate.net/profile/Mythily-Ganesh/publication/362747012_An_Analysis_of_the_Significance_of_Spring_Boot_in_The_Market/links/64c7bc5b3d1a321c1b4cf23a/An-Analysis-of-the-Significance-of-Spring-Boot-in-The-Market.pdf?origin=publication_detail
5	https://apidog.com/blog/rest-api-url-best-practices-examples/
6	https://www.w3schools.com/react/

List of Experiments

1	Design and implement a basic web page using HTML, CSS, and JavaScript that interacts with the user by handling events like button clicks, form submissions, and dynamic content updates. Ensure the page is responsive and functions across different browsers.
2	Develop a simple Single Page Application (SPA) using ReactJS with multiple components. Implement basic state management and event handling to display a dynamic list of items fetched from a mock API. Utilize JSX and Props for managing the UI.
3	Write JavaScript code to handle asynchronous operations using Promises and the Fetch API to communicate with a RESTful API. Display the fetched data on a web page and handle possible errors during the data retrieval process.
4	Optimize a web page for faster loading by implementing techniques like JavaScript minification, image optimization, and lazy loading of media. Analyze the performance of the page before and after optimization using browser developer tools.
5	Create a simple form in ReactJS that allows users to input data. Implement form validation using JavaScript to ensure correct data entry before submission. Use React state management to update the form in real-time.
6	Create a Spring Boot application for a financial institution that needs different configurations for development, testing, and production environments. Manage the configurations using application.yml to handle database connections, logging levels, and external service URLs.
7	Design a REST API for an online booking system where users can book, cancel, and modify reservations. Ensure the API follows REST principles and properly handles different HTTP requests.
8	Build a Spring Boot application that interacts with a relational database to store and retrieve customer feedback for an online service. Optimize the database queries using JPQL and custom queries to retrieve insights such as the most frequent feedback topics.
9	Create a registration system for an educational platform that validates user input such as email, passwords, and age criteria. Ensure that the system handles invalid data gracefully with custom validation messages for the users.

MT23CE04A:: Cloud Computing

Course Type	MC	Semester	6
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Teaching Scheme		Credits		Examination Scheme	
Lecture	3Hr./Week	Lecture	3	CIE Marks	60
Tutorial	--Hr./Week	Tutorial	-	ESE Marks	40
Practical/Studio	0Hr./Week	Practical/Studio	0	Total Marks	100
Total	3Hr./Week	Total	3		

Course Description

This course gives a deep insight into cloud computing principles, models, and services. It includes cloud architecture, virtualization, security, and hands-on practice with major cloud platforms like AWS, Azure, and Google Cloud. Students will learn about cloud storage, security issues, and emerging trends in cloud technology, which include edge computing and serverless architectures.

Course Outcomes

CO No.	Statement
1	Explain the fundamental concepts, characteristics, and benefits of cloud computing.
2	Understanding cloud architecture and security aspects.
3	Demonstrate knowledge of virtualization technologies and their role in cloud computing.
4	Synthesis of different cloud simulators for real time application development.
5	Deployment of applications on cloud platforms .

Mapping of COs to POs and PSOs

CO No	Pos												PSOs				BT L
	P O 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PS O1	PSO 2	PSO 3	PSO 4	
CO1	3																UN
CO2		2															UN
CO3					2												AP
CO4									2								EV
CO5															3		CR

Affinity Level: 1- Slight, 2- Moderate, 3-Substantial, BTL: Bloom's Taxonomy Level

Course Content

Unit No 1	Introduction to Cloud Computing	Hours 9	CO	BTL
	Basics of Distributed and Parallel Computing , Basics of Distributed and Parallel Computing ,Evolution of Cloud Computing, Characteristics of Cloud Computing, Benefits and Limitations of Cloud Computing, Cloud Computing vs. Traditional Computing, Cloud Adoption Challenges, Applications of Cloud Computing		1	UN
Unit No 2	Cloud Architecture and Security	Hours 9	2	AN
	Cloud Computing Architecture ,Virtualization and Its Role in Cloud Computing ,Cloud Service Models: Infrastructure as a Service (IaaS) ,Platform as a Service (PaaS) ,Software as a Service (SaaS) ,Cloud Deployment Models: Public Cloud ,Private Cloud ,Hybrid Cloud ,Community Cloud . Cloud Security Models ,Identity and Access Management (IAM) ,Threats in Cloud Security and Mitigation Strategies.			
Unit No 3	Virtualization and Resource Management	Hours 9	3	UN
	Basics of Virtualization , Types of Virtualizations: Compute, Storage, and Network Hypervisors: Type 1 & Type 2 ,Virtual Machines vs. Containers , Docker and Kubernetes Overview, Resource Allocation and Load Balancing.			
Unit No 4	Cloud Simulators	Hours 9	3	UN
	CloudSim and GreenCloud : Introduction to Simulator, understanding CloudSim simulator, CloudSim Architecture(User code, CloudSim, GridSim, SimJava) Understanding Working platform for CloudSim, Introduction to GreenCloud.			
Unit No 5	Cloud Platforms, Applications, and Emerging Trends	Hours 9	5	AP
	Overview of Cloud Service Providers: AWS, Azure, Google Cloud ,Hands-on with Cloud Services: Compute (EC2, Virtual Machines, Lambda) ,Storage (S3, Blob Storage, Google Cloud Storage) , Databases (DynamoDB, Cloud SQL) Cloud in IoT, AI, and Big Data ,Serverless Computing & Edge Computing Cloud-Native Development and microservices.			

Textbooks

1	<i>Rajkumar Buyya, Christian Vecchiola, S. Thamarai Selvi, "Mastering Cloud Computing", McGraw Hill Education, 2013.</i>
2	<i>Kai Hwang, Geoffrey C. Fox, Jack J. Dongarra, "Distributed and Cloud Computing", Morgan Kaufmann, 2011.</i>
3	<i>"Cloud Computing: A Hands-On Approach" Arshdeep Bahga, Vijay Madisetti 2023</i>

Reference Books / Journal Articles / Weblink

1	<i>"Cloud Computing: Concepts, Technology & Architecture" Thomas Erl, Zaigham Mahmood, and Ricardo Puttini 2013 Pearson Education</i>
2	<i>"Architecting the Cloud: Design Decisions for Cloud Computing Service Models (SaaS, PaaS, and IaaS)" – Michael J. Kavis (2023)</i>

3	"Cloud Computing" Nayan B. Ruparelia 2023 The MIT Press
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BTECCE23H610:: Cloud & Mobile Security

Course Type	Hons	Semester	VI
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Teaching Scheme		Credits		Examination Scheme	
Lecture	3Hr./Week	Lecture	3	CIE Marks	60
Tutorial	--Hr./Week	Tutorial	-	ESE Marks	40
Practical/Studio	0Hr./Week	Practical/Studio	0	Total Marks	100
Total	3Hr./Week	Total	3		

Course Description

This course provides a comprehensive understanding of Cloud and Mobile Security, covering fundamental concepts, security challenges, and industry's best practices. It explores cloud computing models, security threats, data protection techniques, compliance standards, and real-world cloud security implementations. Additionally, it delves into mobile security threats, secure mobile application development, endpoint security, and enterprise mobility management. The course also examines the integration of cloud and mobile security, AI-driven threat detection, and case studies from leading tech enterprises. By the end, students will be equipped to analyze, implement, and integrate security solutions for cloud and mobile environments.

Course Outcomes

CO No.	Statement
1	To understand the fundamentals of cloud computing and mobile security.
2	To explore various security challenges in cloud environments.
3	To study industry-grade cloud security solutions and best practices.
4	To understand mobile security threats and mitigation techniques.
5	To analyze real-world industry implementations of integrated cloud and mobile security

Mapping of COs to POs and PSOs

CO No	POs												PSOs				BTL
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4	
CO1	3																UN
CO2		2															AP
CO3			3														AN
CO4					3												AP
CO5															3		AN

Affinity Level: 1- Slight, 2- Moderate, 3-Substantial, BTL: Bloom's Taxonomy Level

Course Content

Unit No 1	Fundamentals of Cloud Computing	Hours 10	CO	BTL
Introduction to Cloud Computing: Definition, Characteristics, Service Models (IaaS, PaaS, SaaS), Deployment Models: Public, Private, Hybrid, and Community Clouds, Virtualization: Concept, Hypervisors (Type-1 & Type-2), Virtual Machines (VMs), Cloud Computing Architecture: Multi-Tenancy, Resource Allocation, Scalability, Emerging Trends: Edge Computing, Fog Computing, Serverless Computing			1	UN
Unit No 2	Security in Cloud Computing	Hours 10	2	AP
Cloud Security Threats: Data breaches, Insider Attacks, DDoS, API Security Security Solutions: Identity and Access Management (IAM), Zero Trust Security Model, Cloud Data Security: Data Encryption (AES, RSA), Secure Data Storage, Tokenization, Cloud Compliance and Governance: GDPR, HIPAA, ISO 27001 Case Studies: Security Measures in AWS, Google Cloud, and Microsoft Azure				
Unit No 3	Industry Security Practices in Cloud Computing	Hours 10	3	AN
Security Frameworks: NIST, CIS Controls, Cloud Security Alliance (CSA) Cloud Security Tools: SIEM, Firewalls (WAF), Intrusion Detection & Prevention (IDS/IPS), Secure DevOps (DevSecOps): CI/CD Pipeline Security, Container Security (Docker, Kubernetes), Incident Response & Forensics in Cloud: Logging, Monitoring, Threat Intelligence, Industry Case Studies: Cloud Security Implementations in Banking, Healthcare, and IT				
Unit No 4	Fundamentals of Mobile Security	Hours 8	4	AP
Mobile Device Security: Android vs. iOS Security Architecture, Mobile Threats: Malware, Rooting/Jailbreaking, Phishing Attacks, Mobile Application Security: OWASP Mobile Top 10 Vulnerabilities, Secure Mobile Communication: VPN, Mobile Data Encryption, Mobile VPNs, Mobile Device Management (MDM) and Endpoint Security.				
Unit No 5	Industry Security Using Cloud & Mobile Security	Hours 7	5	AN
Integration of Cloud & Mobile Security: Secure APIs, Cloud-Based Mobile Security Solutions, Enterprise Mobile Security Strategies: BYOD (Bring Your Own Device), EMM (Enterprise Mobility Management), Security in Edge & IoT Devices: Secure Edge Computing, IoT Security Frameworks, AI and ML in Cloud & Mobile Security: Threat Detection, Anomaly Detection, Automated Incident Response, Case Studies: Real-World Enterprise Security Solutions (Google, Apple, Amazon, Microsoft)				

Textbooks

1.	<i>Cloud Security Handbook – Eyal Estrin, Packt Publishing</i>
2.	<i>Mobile Security & Privacy – Man Ho Au & Raymond Choo, Syngress</i>
3.	<i>Cloud Security & Privacy – Tim Mather, Subra Kumaraswamy, O'Reilly</i>
4.	<i>Cloud Computing Security – John Rittinghouse, James Ransome, CRC Press</i>

